High-Order Triplet CRF-PCANet for Unsupervised Segmentation of Nonstationary SAR Image

Peng Zhang, *Member, IEEE*, Mohamed El Yazid Boudaren, Yinyin Jiang, Wanying Song, *Member, IEEE*, Beibei Li, Ming Li, and Yan Wu, *Member, IEEE*

***Abstract*— Conditional random fields (CRFs) model is suitable for image segmentation because it can capture the dependencies of observed data and incorporate the spatial correlations into the segmentation process. In this article, to deal with the segmentation of nonstationary synthetic aperture radar (SAR) image, we combine the modeling power of the CRF model with the representation-learning ability of principal component analysis network (PCANet), and thus propose a high-order triplet CRF model based on PCANet (HOTCRF-PCANet). HOTCRF-PCANet introduces an auxiliary field to explicitly regulate nonstationary label structure patterns. Under the guidance of this auxiliary field, HOTCRF-PCANet defines a discrete quadrilateral nonstationary Markov fields model, and thus considers both the nonstationary property of image and high-order label interactions. In addition, guided by the auxiliary field, HOTCRF-PCANet proposes to use a product-of-expert (POE) potential to enforce the regions' labeling consistency for pixels within the weak-structured region. To automatically learn rich feature representations, HOTCRF-PCANet modifies PCANet into an unsupervised mode, i.e., unsupervised PCANet (UPCANet), and constructs an UPCANet-based unary potential to effectively predict the local class probability. The effectiveness of HOTCRF-PCANet is demonstrated by the application to the unsupervised segmentation of the simulated images and real SAR images.**

***Index Terms*— Conditional random fields (CRFs), deep learning, high-order label interactions, nonstationary synthetic aperture radar (SAR) image, unsupervised segmentation.**

I. Introduction

SYNTHETIC aperture radar (SAR) image segmentation is the fundamental step to the image interpretation, and promotes the applications of SAR system in many fields, such as geological exploration and environmental monitoring. Due to the inherent speckle, SAR image segmentation is generally acknowledged as the challenging work, and a wide variety of methods have been developed recently [1]–[7].

Conditional random fields (CRFs) model is able to capture the dependencies of observed data and incorporate the spatial correlations into the segmentation process [8]. Consequently, the CRF model is acknowledged as a powerful model for image segmentation and has been extended to several variants, such as multiple CRF model [9], hierarchical CRF model [10], triplet CRF model [11], and so on [12]–[14]. Based on Bayesian fusion, multiple CRF models can incorporate multiple features effectively [9]. The hierarchical CRF model can effectively exploit the global and local image information including the contextual structures and textural features [10]. Triplet CRFs model considers the nonstationarity of image from the perspective of texton appearances, and thus captures the distribution differences of the congeneric image features in different stationary parts in the labeling [11]. Although the CRF models discussed above are effective in image segmentation and classification, there are two potential improvements of the CRF model as follows.

First, these CRF models discussed above have a limited ability to capture the high-order label structure information due to only the pairwise potential being constructed. In fact, as it is known, a pixel at some site is highly dependent on its spatial neighbors. For instance, in the region of man-made structures, the spatially adjoining sites that consist of the lines or edges follow some underlying organization rules. Consequently, it is essential to capture multidimensional label interactions in a high-order neighborhood system, rather than pairwise one in a 4-neighborhood system. To model the high-order label structures, Stewart *et al.* [12] propose a higher-order CRF model and use a set of local filters to model the complex label interactions. However, the local filters need to be trained using a large number of training data, which is not feasible for unsupervised SAR image segmentation. In addition, He *et al.* [13] propose a multiscale CRF model and define the posterior distribution as a product of three conditional distributions, and thus capture the multiscale information including the image features and label structures. But a large amount of training data are also needed to learn the model parameters in practice. A high-order CRF using robust P^n Potts model proposed by Kohli *et al.* [14] enforces label consistency in image regions and has achieved promising results in natural image segmentation. But its model parameters need to be trained by cross-validation, which is not suitable for unsupervised SAR image

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Peng Zhang, Yinyin Jiang, Beibei Li, and Ming Li are with the National Key Laboratory of Radar Signal Processing, Collaborative Innovation Center of Information Sensing and Understanding, Xidian University, Xi'an 710071, China (e-mail: pzhang@xidian.edu.cn).

Mohamed El Yazid Boudaren is with the Ecole Militaire Polytechnique, Algiers 16111, Algeria.

Wanying Song and Yan Wu are with the Remote Sensing Image Processing and Fusion Group, School of Electronic Engineering, Xidian University, Xi'an 710071, China.

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segmentation. To capture the high-order label dependencies in an unsupervised way, Zhang *et al.* [15] propose a high-order CRF model based on product-of-expert (POE). In addition, higher order neighborhood-based triplet Markov fields (HNTMFs) model proposed recently enhances the prior modeling ability to describe the label interaction [16]. On the other hand, region CRF models proposed based on superpixels incorporate larger scale contextual information in labeling and thus successfully improve the homogeneous regions' labeling consistency [17], [18]. However, these random field models above generally focus on the label consistency and thus have a limited ability to model the complex label interactions.

In addition, CRF models generally extract handcrafted features, such as histogram measures and semivariagrams, to predict the class probability [15], [19]. But due to speckle and high dynamics of SAR image, handcrafted features are not quite effective in practice. Consequently, for SAR image segmentation, statistical distributions, such as Gamma distribution [7] and generalized Gamma (GGamma) distribution [10], are usually used in CRF to model SAR scattering statistics and thus offer statistical properties to discriminate the classes of SAR image [10], [15]. Recently, deep neural networks (DNNs) prove to excel at many computer vision problems due to the ability of learning-rich feature representations from data [19]–[24]. DNNs learn hierarchical contextual features from the image by convolutional kernels, and the learned deep features have exhibited great generalization ability over the handcrafted features [19]–[21]. Afterward, CRF models are integrated with DNNs, and have achieved promising results in segmentation and classification [23], [25], [26]. Consequently, it is feasible to use DNNs to extract rich and deep features from SAR images. But in practice, DNNs generally need many training samples to learn the convolution filters, weight vector, and bias, which limits the application of DNNs to unsupervised SAR image segmentation. The recent principal component analysis network (PCANet) is also a kind of deep learning networks [27]. It first uses principal component analysis to learn convolution filters, and then utilizes binary hashing to code the hierarchical features extracted from the images. PCANet is a very simple but highly competitive deep learning network, and has been applied to face recognition and image change detection [27], [28]. To our knowledge, it has not yet been used for unsupervised SAR image segmentation.

Motivated by the discussion above, in this article, we propose a high-order triplet CRF model based on PCANet (HOTCRF-PCANet) for the unsupervised segmentation of nonstationary SAR images. HOTCRF-PCANet contains the following two contributions to unsupervised nonstationary SAR image segmentation:

1) HOTCRF-PCANet captures the high-order label interactions effectively in an unsupervised way.
   First, HOTCRF-PCANet defines the nonstationary property of image as the nonstationarity of the label structure pattern, which implies that a nonstationary image consists of weak-structured and intense-structured regions with different kinds of label interaction. Then, HOTCRF-PCANet introduces an auxiliary field to segment the image into weak-structured and intense-structured regions and thus manages the different label structure patterns.
   Guided by the auxiliary field which manages the nonstationarity of label structure patterns, HOTCRF-PCANet proposes a discrete quadrilateral nonstationary Markov fields (DQNMFs) to compute the multidimensional label probability, and thus constructs a high-order DQNMF potential to capture the multidimensional label interactions within a high-order neighborhood system. "Quadrilateral" means that four unilateral Markovian constraints on the lattice with respect to four orientations, including upper-right, upper-left, down-right, and down-left corners, are enforced, as given in (7) in Section IV-A. In addition, guided by the auxiliary field, for the pixels in the weak-structured region, HOTCRF-PCANet proposes to enforce label consistency using a POE potential. In this way, HOTCRF-PCANet can capture the high-order label structure accurately, and is robust against speckle.
2) HOTCRF-PCANet combines the label-interaction modeling power of CRF with the representation-learning ability of PCANet.
   To enhance the segmentation performance, a feasible way is to integrate a deep learning network into the CRF framework. Consequently, in this article, HOTCRF-PCANet modifies PCANet into an unsupervised mode, named as unsupervised PCANet (UPCANet). Thus HOTCRF-PCANet automatically learns rich features from SAR image and constructs an UPCANet-based unary potential to measure the class probability. To our knowledge, it is the first time to modify this highly competitive convolutional network to fit unsupervised SAR image segmentation. In this way, both high-order label interactions and deep features can be captured by HOTCRF-PCANet, which can enhance the labeling ability effectively.

The structure of this article is as follows. Section II recalls the CRF model. Section III presents the nonstationary property of label structure pattern. Section IV presents the proposed HOTCRF-PCANet. Section V discusses the parameter estimation. The experimental results are presented in Section VI. Section VII gives the concluding remarks.

II. CRFs Model

CRF model is proposed by Kumar and Hebert [8] for image labeling. As a discriminative model, CRF model extracts the image features from an arbitrary patch centered at site (i, j) to construct the unary potential, and thus directly models the posterior distribution as a Gibbs distribution. Consequently, CRF model is able to capture dependencies of the observed data, and has proven to be suitable to analyze the images with textural structures [8].

Let S denote a finite set of sites on finite rectangular lattices with a neighborhood system defined on it. Assume that $X = (X_{i,j})_{(i,j)\in S}$ and $Y = (Y_{i,j})_{(i,j)\in S}$ are two random fields defined on S. Let $\Omega = \{e_1, e_2, \ldots, e_L\}$ be a finite set with L elements, each being represented by a binary vector

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of length L with one component being 1, all others being 0. Then in the context of image segmentation using random fields model, $X = (X_{i,j})_{(i,j)\in S}$ is the label field with each $X_{i,j}$ taking values in Ω and representing the class to which the site belongs, and $Y = (Y_{i,j})_{(i,j)\in S}$ is the observed field which can be seen as a noisy version of $X = (X_{i,j})_{(i,j)\in S}$, taking values in $\mathbb{R}$. The task of image segmentation is to recover the label field $X = (X_{i,j})_{(i,j)\in S}$ from the observed field $Y = (Y_{i,j})_{(i,j)\in S}$. $x = (x_{i,j})_{(i,j)\in S}$ and $y = (y_{i,j})_{(i,j)\in S}$ denote the realizations of $X = (X_{i,j})_{(i,j)\in S}$ and $Y = (Y_{i,j})_{(i,j)\in S}$, respectively.

In the CRF model, the hidden random variables $X = (X_{i,j})_{(i,j)\in S}$ are Markovian conditioned on observations $Y = (Y_{i,j})_{(i,j)\in S}$. Then using the Hammersley–Clifford theorem [29] and only considering the pairwise clique potentials, the posterior distribution of the CRF model can be expressed as

$$p(x|y) = \frac{1}{Z_{\mathrm{CRF}}} \exp\left[\sum_{(i,j)\in S} A_{i,j}(x_{i,j}, h_{i,j}(y)) + \sum_{(i,j)\in S}\sum_{(i',j')\in \mathrm{N}_{(i,j)}} I(x_{i,j}, x_{i',j'}, \mu_{(i,j),(i',j')}(y))\right] \tag{1}$$

where Z_{CRF} denotes the partition function, $\mathrm{N}_{(i,j)}$ denotes the 4-neighborhood of site (i, j), $h_{i,j}(y)$ is the feature vector at site (i, j) that spans a feature space for the discriminative classifier, $\mu_{i,j,i',j'}(y)$ denotes the feature for a pair of sites (i, j) and (i', j'), which is used to model the pair-site label interaction, $A_{i,j}(x_{i,j}, h_{i,j}(y))$ is the unary potential of the CRF model, and is defined as

$$A_{i,j}(x_{i,j}, h_{i,j}(y)) = \sum_{e_l\in\Omega} \delta(x_{i,j}, e_l) \log P'(x_{i,j} = e_l|h_{i,j}(y)) \tag{2}$$

where $P'(x_{i,j} = e_l|h_{i,j}(y))$ is a domain-specific discriminative classifier that can be traditionally defined by support vector machine (SVM) and logistic classifier, such as nominal logistic regression model (NLRM) [15], [30]. $I(x_{i,j}, x_{i',j'}, \mu_{(i,j),(i',j')}(y))$ denotes the pairwise potential, and can be defined as

$$\begin{aligned} I(x_{i,j}, x_{i',j'}, \mu_{(i,j),(i',j')}(y)) &= \sum_{\{(i,j),(i',j')\}\in N_H} \alpha_H(2\delta(x_{i,j}, x_{i',j'}) - 1) \times \mu_{(i,j),(i',j')}(y)) \\ &+ \sum_{\{(i,j),(i',j')\}\in N_V} \alpha_V(2\delta(x_{i,j}, x_{i',j'}) - 1) \times \mu_{(i,j),(i',j')}(y)) \end{aligned} \tag{3}$$

where N_H and N_V are the sets of the horizontal and vertical pairwise cliques, respectively, $\{\alpha_H, \alpha_V\}$ are the interaction parameters that denote the label structure pattern, and $\mu_{(i,j),(i',j')}(y)$ denotes the edge strength that is utilized to model the pair-site label interaction. $\delta(x_{i,j}, x_{i',j'})$ verifies $\delta(x_{i,j}, x_{i',j'}) = 1$ for $x_{i,j} = x_{i',j'}$, and $\delta(x_{i,j}, x_{i',j'}) = 0$ for $x_{i,j} \neq x_{i',j'}$.

CRF model captures the dependencies of observed data and models the spatial label interaction. Although CRF model is effective in image segmentation, classification, and change detection, there are some potential improvements of the original CRF model: 1) CRF model does not take into account the nonstationarity of label structure pattern, and implies that the label structure pattern is always the same, which facilitates to model the spatial label interaction. That is to say, only one set of model parameters $\{\alpha_H, \alpha_V\}$ is used to model the label interaction. However, when dealing with nonstationary SAR image, the potential different label structure patterns should be considered; 2) Second, due to only the pairwise potential being constructed, it has a limited ability to capture high-order label interactions. The pairwise potential only considers the pair-site label interaction within a 4-neighborhood. Then, the label dependencies are limited within the pairwise cliques, which is prone to implement the label inference in practice. However, the pairwise potential cannot effectively represent complex label interactions, which limits its segmentation performance in practice; 3) In addition, CRF model traditionally extracts handcrafted features, such as histogram measures and semivariograms, to measure the probability of the hidden random variable taking some label. But due to speckle and high dynamics of SAR image, the generalization ability of handcrafted features is usually inadequate, and they are not quite effective in practice. Consequently, effective feature representation should be taken into account.

III. Nonstationary Property of Label Structure Pattern

In image segmentation, the nonstationary property of image was first considered by Benboudjema and Pieczynski in triplet Markov fields (TMFs) model [31]. Afterward, the nonstationarity of image is discussed by several works and more explicitly defined [11], [32]–[36]. For instance, the nonstationarities of noise distributions are taken into account and each class is assigned a set of noise densities, which is more reasonable in practice [36]. Additionally, the nonstationarity of image is considered from the perspective of texton appearances in the modeling of image features [11].

In this article, the nonstationarity of image explicitly expresses that there are several kinds of label structure pattern in an image. Concretely, in an image with complex scene, we usually have weak-structured regions, and intense-structured regions. Weak-structured region has weak textural representation, and corresponds to homogeneous region in the segmentation. As shown in Fig. 1, Region1 is a weak-structured region. Intense-structured regions, by contrast, have strong textural representations, and can be further divided according to the different potential label structure patterns. For instance, Region2 and Region3 are intense-structured regions, in which the horizontal structure and vertical structure are the dominant, respectively.

In the segmentation, the potential different label structure patterns should be considered. Label interaction means that a pixel belonging to some class is highly dependent on its neighbors. Generally, when the label structure patterns are different,

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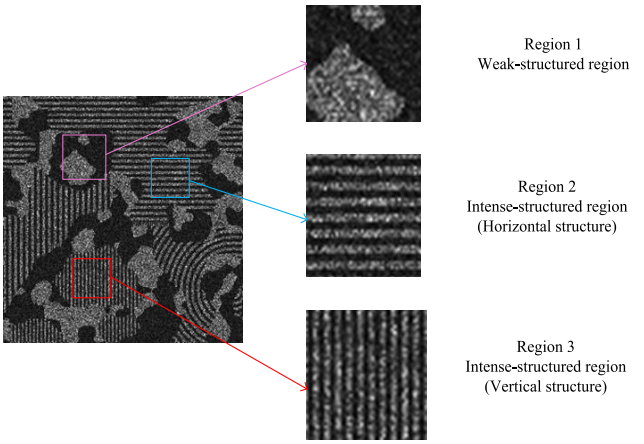

Fig. 1. Nonstationary image and potential different label structure patterns.

the corresponding label interactions are different. Specifically, in the horizontal-structured Region2, a pixel belonging to some class is more highly dependent on its horizontal neighbors, and the horizontal label interaction should be emphasized in the segmentation. In the vertical-structured Region3, a pixel belonging to some class is more highly dependent on its vertical neighbors, and we should pay more attention to the vertical label interaction. And in Region 1, the label consistency should be enforced.

In addition, a nonstationary SAR image with complex scene could be generally divided into weak-structured and intense-structured regions. The reason is as follows. A nonstationary SAR image usually consists of vegetable region and artificial building zone. In the vegetable region, the pixels within a local window are usually highly dependent, and the values and labels of these pixels do not change greatly. That is to say, the pixels have weak textural representation. Consequently, the vegetable region could be considered as the weak-structured region. In the artificial building zone, the spatially adjoining sites consist of the lines or edges of man-made buildings, and thus the pixels' intensities and labels within a local lattice follow some underlying organization rules. That is to say, the pixels in this region have intense textural representation. That is why the artificial building zone could be considered as the intense-structured region.

Then, in the context of random fields model, we define an auxiliary field $U = (U_{i,j})_{(i,j)\in S}$ to regulate these different structure patterns. Each $U_{i,j}$ takes its value in $\Lambda = \{g_1, g_2, \ldots, g_M\}$, where M denotes the number of the potential label structure patterns. In practice, $U_{i,j} = g_1$ represents that (i, j) is within a weak-structured region, and $U_{i,j} = g_m, 2 \le m \le M$ represents that (i, j) is within an intense-structured region. Each structure pattern corresponds to a kind of label interaction. Then, we have the interaction parameters $\alpha = \{\alpha_H^1, \alpha_{mH}^2, \alpha_V^1, \alpha_{mV}^2\}, 1 \le m \le M$ to model the label interaction in the modeling of multidimensional label interactions within the high-order neighborhood system, which will be discussed in Section IV.

To partition the auxiliary field, we use the nonstationary anisotropic Gaussian kernel (NAGK). NAGK is related with the local autocovariance of SAR image and models the spatial variability and autocorrelation [37]. In practice, NAGK is oriented with the angle ω and characterized by the spatial standard deviations $\{\sigma_h, \sigma_v\}$, and the local autocovariance of SAR image can be expressed as a function of the NAGK parameters $\{\omega, \sigma_h, \sigma_v\}$ as follows: $AC_T(d) = \sigma_T^2 \exp(-d^T \Sigma_T^{-1} d)$, where d denotes the 2-D displacement vector between the two sites, σ_T^2 is the $(0, 0)$ coefficient of AC_T, and Σ_T can be decomposed as $\Sigma_T = M_\omega^T \Xi M_\omega$, where $M_\omega = \begin{bmatrix} \cos\omega & -\sin\omega \\ \sin\omega & \cos\omega \end{bmatrix}$, and $\Xi = \begin{bmatrix} (2\sigma_h)^2 & 0 \\ 0 & (2\sigma_v)^2 \end{bmatrix}$. The NAGK parameters $\{\omega, \sigma_h, \sigma_v\}$ can be estimated from the local autocovariance of SAR image by means of the geometric moments [37]. Given the NAGK feature vector of site (i, j), $f_{\text{NAGK}}^{i,j} = \{\omega^{i,j}, \sigma_h^{i,j}, \sigma_v^{i,j}\}_{(i,j)\in S}$, we can initialize the auxiliary field by fuzzy C-means (FCM). The principle of standard FCM algorithm is to minimize the fuzzy objective function, which is given by

$$J_{\text{FCM}}^{\phi} = \sum_{(i,j)\in S} \sum_{m=1}^{M} \left[r_{m(i,j)}^{U}\right]^{\phi} \left\| f_{\text{NAGK}}^{i,j} - c_m \right\|^2 \tag{4}$$

where $r_{m(i,j)}^{U}$ denotes the degree of membership of $f_{\text{NAGK}}^{i,j}$ in the mth cluster, and ϕ is a weighting exponent on each fuzzy membership value $r_{m(i,j)}^{U}$, c_m is the center of the mth cluster. Then, the auxiliary field $U = (U_{i,j})_{(i,j)\in S}$ can be initialized according to the degree of membership.

IV. Proposed HOTCRF-PCANet

We propose a high-order triplet CRF-PCANet for nonstationary SAR image segmentation in this section, and take the following three points into account:

1) HOTCRF-PCANet introduces an auxiliary field to consider the different label structure patterns within the intense-structured and weak-structured regions.
2) To model the high-order label interactions, we propose a DQNMFs model and construct a high-order DQNMF potential. In addition, for pixels within the weak-structured region, we construct a POE potential to enforce the regions' labeling consistency under the guidance of the auxiliary field. In this way, the proposed HOTCRF-PCANet considers both the nonstationary property and high-order label interactions in an unsupervised way.
3) Finally, PCANet is modified into an unsupervised mode, named as UPCANet, to fit unsupervised segmentation. UPCANet automatically extracts rich features from SAR images, and then an UPCANet-based unary potential is constructed to predict the class probability.

We define the following notations and discuss how to construct the proposed HOTCRF-PCANet in detail. $S_{\text{HO}_{(i',j')}}$ is an overlapping high-order neighborhood that is a $n \times n$ rectangular lattice of sites with the centered site being (i', j'). $(\mathbf{X}_{\mathbf{HO}_{(i',j')}}, \mathbf{U}_{\mathbf{HO}_{(i',j')}})$ denote the high-order hidden random variables defined on $S_{\text{HO}_{(i',j')}}$, and $(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}})$ denote the realization of $(\mathbf{X}_{\mathbf{HO}_{(i',j')}}, \mathbf{U}_{\mathbf{HO}_{(i',j')}})$. Considering the three factors above, we define the posterior distribution of

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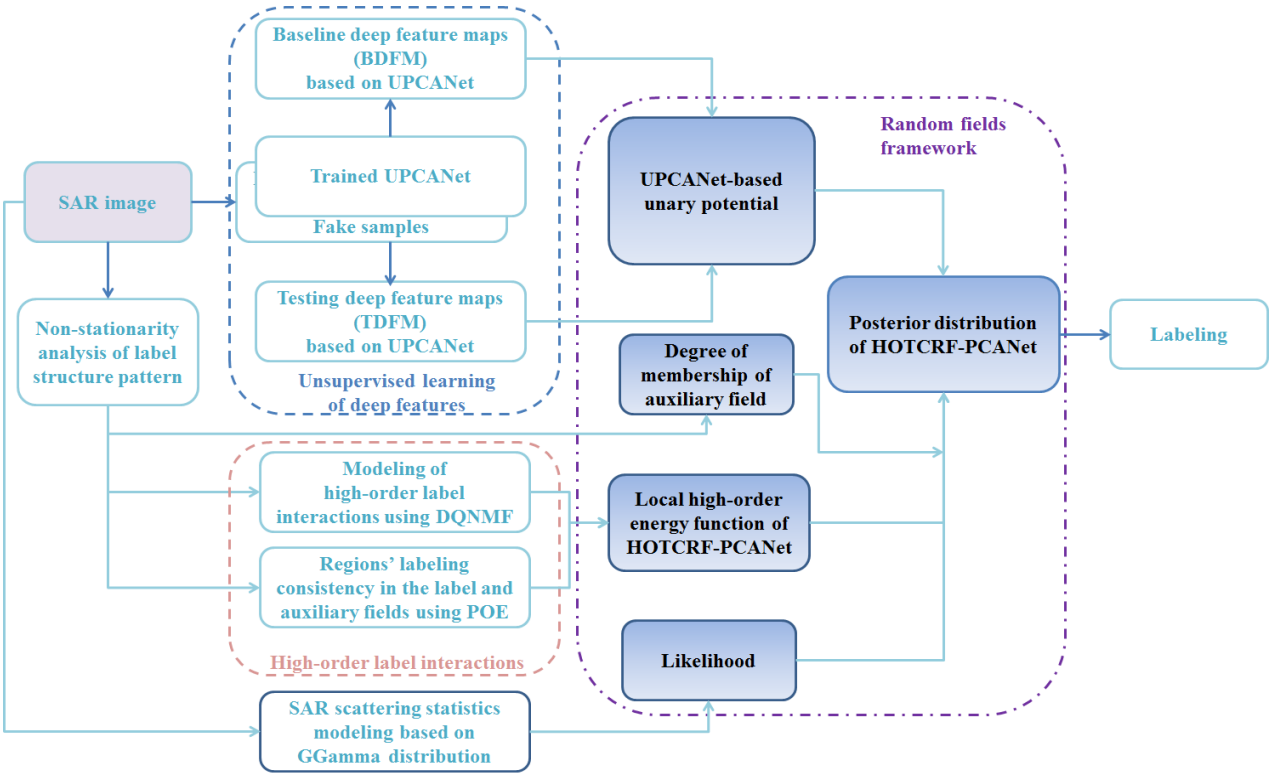

Fig. 2. Segmentation framework using the proposed HOTCRF-PCANet.

HOTCRF-PCANet as

$$p(x,u|y) = \frac{1}{Z}\exp\left\{\sum_{(i',j')\in S}\left[E(\mathbf{x}_{\mathbf{HO}_{(i',j')}},\mathbf{u}_{\mathbf{HO}_{(i',j')}}) + \log A^{\mathrm{UPCANet}}(x_{i',j'}|y) + \log p(y_{i',j'}|x_{i',j'}) + \log\left(r^{U}_{i',j'}\right)\right]\right\} \quad (5)$$

where Z denotes the partition function in the framework of random fields model, $E(\mathbf{x}_{\mathbf{HO}_{(i',j')}},\mathbf{u}_{\mathbf{HO}_{(i',j')}})$ is the local high-order energy function that models the high-order label interactions under the guidance of the auxiliary field U, and $A^{\mathrm{UPCANet}}(x_{i',j'}|y)$ is the UPCANet-based unary potential that predicts the local class probability using deep features. The likelihood $p(y_{i',j'}|x_{i',j'})$ models the statistics of SAR data and describes the variability within the consistent class, and $r^{U}_{i',j'}$ denotes the degree of membership of $f^{i,j}_{\mathrm{NAGK}}$ in the segmentation of the auxiliary field, as discussed in Section III. An overview of the segmentation framework using the proposed HOTCRF-PCANet is given in Fig. 2. In the following, we will discuss how to construct the posterior distribution in detail.

A. Construction of High-Order Energy Function

HOTCRF-PCANet considers both the nonstationary property and high-order label interactions by the construction of high-order energy function. When constructing the local high-order energy function to model the multidimensional label interactions, we first propose a DQNMFs model and thus construct a high-order DQNMF potential to model the multidimensional label interactions. Second, to enforce the regions' labeling consistency of HOTCRF-PCANet, a POE potential for the pixels within the weak-structured region is incorporated into the high-order energy function. Then, the local high-order energy function is defined as

$$E(\mathbf{x}_{\mathbf{HO}_{(i',j')}},\mathbf{u}_{\mathbf{HO}_{(i',j')}}) = E^{\mathrm{DQNMF}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}},\mathbf{u}_{\mathbf{HO}_{(i',j')}}) + E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}})\cdot\delta(u_{i',j'},g_1) + E^{\mathrm{POE}}(\mathbf{u}_{\mathbf{HO}_{(i',j')}}) \quad (6)$$

where $E^{\mathrm{DQNMF}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}},\mathbf{u}_{\mathbf{HO}_{(i',j')}})$ is the high-order DQNMF potential that models the multidimensional label interactions, $E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}})\cdot\delta(u_{i',j'},g_1)$ is the POE potential designed for the labeling consistency for pixels within the weak-structured region, $E^{\mathrm{POE}}(\mathbf{u}_{\mathbf{HO}_{(i',j')}})$ is the POE potential designed for the labeling consistency in the auxiliary field, which is constructed in the same way as $E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}})$. In this way, the proposed HOTCRF-PCANet considers both the nonstationary property and high-order label interactions.

1) High-Order DQNMF Potential: First, to construct a high-order DQNMF potential and model the multidimensional label interactions, we propose a DQNMF based on quadrilateral Markov fields (QMRFs). QMRF model was originally proposed to compute the image spatial entropy of continuous-valued image, and has proven to be a promising tool for image fusion and registration, and difference image of multi-temporal data [38], [39]. In this section, we propose DQNMF based on QMRF for modeling high-order label structure in image segmentation. Concretely, DQNMF regulates the label structure

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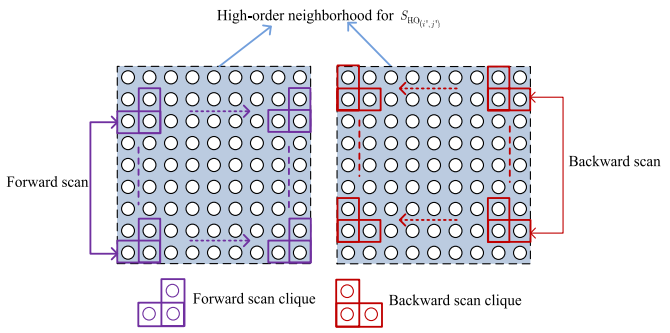

Fig. 3. High-order neighborhood system.

patterns by the auxiliary field and models the multidimensional label interactions within the high-order neighborhood.

DQNMF is defined as follows. $X = (X_{i,j})_{(i,j)\in S}$ is the label field and $U = (U_{i,j})_{(i,j)\in S}$ is the auxiliary field that regulates different label structure patterns. Random fields $(X_{i,j}, U_{i,j})_{(i,j)\in S}$ are DQNMF on a finite rectangular lattice if the four unilateral constraints are enforced

$$\begin{aligned} p\left(x_{i,j}, u_{i,j} \middle| Z_{i,j}^{ul}\right) &= p(x_{i,j}, u_{i,j} | x_{i-1,j}, u_{i-1,j}, x_{i,j-1}, u_{i,j-1}) \\ p\left(x_{i,j}, u_{i,j} \middle| Z_{i,j}^{dr}\right) &= p(x_{i,j}, u_{i,j} | x_{i+1,j}, u_{i+1,j}, x_{i,j+1}, u_{i,j+1}) \\ p\left(x_{i,j}, u_{i,j} \middle| Z_{i,j}^{ur}\right) &= p(x_{i,j}, u_{i,j} | x_{i-1,j}, u_{i-1,j}, x_{i,j+1}, u_{i,j+1}) \\ p\left(x_{i,j}, u_{i,j} \middle| Z_{i,j}^{dl}\right) &= p(x_{i,j}, u_{i,j} | x_{i+1,j}, u_{i+1,j}, x_{i,j-1}, u_{i,j-1}) \end{aligned} \tag{7}$$

where $Z_{i,j}^{ul} = \{(x_{i'',j''}, u_{i'',j''}) | i'' < i \text{ or } j'' < j\}$, $Z_{i,j}^{dr} = \{(x_{i'',j''}, u_{i'',j''}) | i'' > i \text{ or } j'' > j\}$, $Z_{i,j}^{ur} = \{(x_{i'',j''}, u_{i'',j''}) | i'' < i \text{ or } j'' > j\}$, $Z_{i,j}^{dl} = \{(x_{i'',j''}, u_{i'',j''}) | i'' > i \text{ or } j'' < j\}$ are the predecessors of $(x_{i,j}, u_{i,j})$.

Then given the overlapping high-order neighborhood $S_{\mathrm{HO}_{(i',j')}}$, using Theorem 1 in the Appendix, we can derive the multidimensional probability as follows:

$$\begin{aligned} &p(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}}) \\ &= \prod_{i=1}^{n'} \prod_{j=1}^{n} p(x_{2i-1,j}, u_{2i-1,j} | x_{2i-2,j}, u_{2i-2,j}, x_{2i-1,j-1}, u_{2i-1,j-1}) \\ &\quad \cdot p(x_{2i,n-j+1}, u_{2i,n-j+1} | x_{2i-1,n-j+1}, u_{2i-1,n-j+1}, \\ &\qquad x_{2i,n-j+2}, u_{2i,n-j+2}) \end{aligned} \tag{8}$$

where n is set to be odd and $n' = (n-1)/2+1$, $(2i-1, j)$, $(2i-2, j)$, $(2i-1, j-1)$, $(2i, n-j+1)$, $(2i-1, n-j+1)$ and $(2i, n-j+2)$ denote the site of the random variables.

According to (8), the multidimensional probability of DQNMF predicts the probability of the hidden random variable at site (i', j') taking some label, and can be derived from a zigzag scan on $S_{\mathrm{HO}_{(i',j')}}$ as shown in Figs. 3 and 4. In this way, we derive a multidimensional probability which consists of a set of conditional probabilities on low-order cliques. Thus, the proposed DQNMF can decompose the high-order label structure into the structure on low-order cliques, consisting of horizontal pairwise cliques and vertical pairwise cliques, which facilitates the modeling of the multidimensional label interactions.

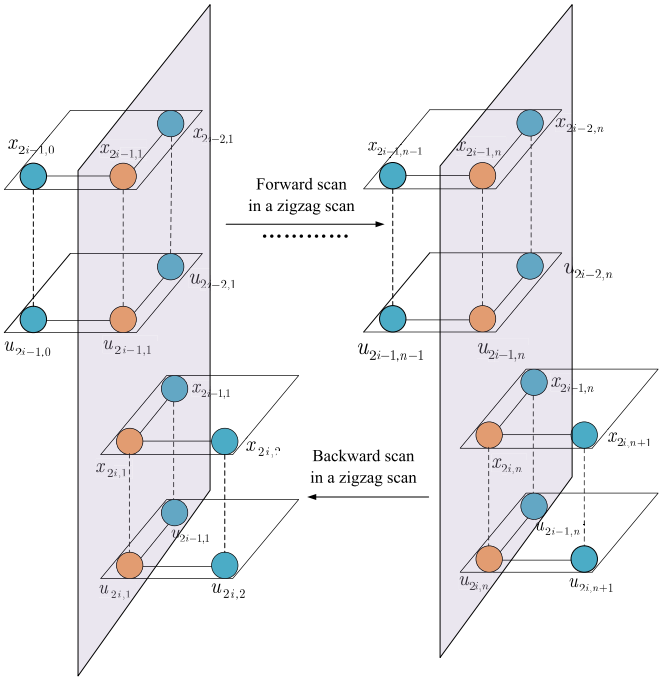

Fig. 4. Computation of the high-order probability of DQNMF through zigzag scans in the high-order neighborhood system.

Using the Potts model [40], the conditional probability in (8) can be defined as

$$\begin{aligned} &p(x_{2i-1,j}, u_{2i-1,j} | x_{2i-2,j}, u_{2i-2,j}, x_{2i-1,j-1}, u_{2i-1,j-1}) \\ &= \frac{1}{Z_{xu}^{2i-1,j}} \exp[E(x_{2i-1,j}, u_{2i-1,j}, x_{2i-2,j}, u_{2i-2,j}) \\ &\qquad + E(x_{2i-1,j}, u_{2i-1,j}, x_{2i-1,j-1}, u_{2i-1,j-1})] \end{aligned} \tag{9-1}$$

$$\begin{aligned} &p(x_{2i,n-j+1}, u_{2i,n-j+1} | x_{2i-1,n-j+1}, u_{2i-1,n-j+1}, \\ &\quad x_{2i,n-j+2}, u_{2i,n-j+2}) \\ &= \frac{1}{Z_{xu}^{2i,n-j+1}} \exp[E(x_{2i,n-j+1}, u_{2i,n-j+1}, x_{2i-1,n-j+1}, u_{2i-1,n-j+1}) \\ &\qquad + E(x_{2i,n-j+1}, u_{2i,n-j+1}, x_{2i,n-j+2}, u_{2i,n-j+2})] \end{aligned} \tag{9-2}$$

where $E(\cdot)$ denotes the pairwise clique potential, and is defined by the Potts model

$$\begin{aligned} &E(x_{i,j}, u_{i,j}, x_{i',j'}, u_{i',j'}) \\ &= \sum_{\{(i,j),(i',j')\}\in \mathrm{N}_H} [2\delta(x_{i,j}, x_{i',j'}) - 1] \\ &\quad \times \begin{bmatrix} \alpha_H^1 (1 - \delta(u_{i,j}, u_{i',j'})) + \alpha_{1H}^2 \delta^*(u_{i,j}, u_{i',j'}, g_1) \\ + \alpha_{2H}^2 \delta^*(u_{i,j}, u_{i',j'}, g_2) + \cdots + \alpha_{MH}^2 \delta^*(u_{i,j}, u_{i',j'}, g_M) \end{bmatrix} \\ &\quad + \sum_{\{(i,j),(i',j')\}\in \mathrm{N}_V} [2\delta(x_{i,j}, x_{i',j'}) - 1] \\ &\quad \times \begin{bmatrix} \alpha_V^1 (1 - \delta(u_{i,j}, u_{i',j'})) + \alpha_{1V}^2 \delta^*(u_{i,j}, u_{i',j'}, g_1) \\ + \alpha_{2V}^2 \delta^*(u_{i,j}, u_{i',j'}, g_2) + \cdots + \alpha_{MV}^2 \delta^*(u_{i,j}, u_{i',j'}, g_M) \end{bmatrix} \end{aligned} \tag{10}$$

where N_H and N_V are the sets of the horizontal and vertical pairwise cliques, respectively, $\alpha = \{\alpha_H^1, \alpha_V^1, \alpha_{mH}^2, \alpha_{mV}^2\}$, $1 \le m \le M$ denotes the interaction parameters in which

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$\{\alpha_{mH}^2, \alpha_{mV}^2\}$ model the label interaction in stationarity m, $\delta(x_{i,j}, x_{i',j'})$ verifies $\delta(x_{i,j}, x_{i',j'}) = 1$ for $x_{i,j} = x_{i',j'}$, and $\delta(x_{i,j}, x_{i',j'}) = 0$ for $x_{i,j} \neq x_{i',j'}$, $\delta^*(u_{i,j}, u_{i',j'}, a) = 1$ for $u_{i,j} = u_{i',j'} = a$, and $\delta^*(u_{i,j}, u_{i',j'}, a) = 0$ otherwise, $Z_{xu}^{2i-1,j}$ and $Z_{xu}^{2i,n-j+1}$ denote the normalizing factor.

Then, the multidimensional probability is reformulated as

$$\begin{aligned} & p(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}}) \\ &= \prod_{i=1}^{n'} \prod_{j=1}^{n} \Bigg\{ \frac{1}{Z_{xu}^{2i-1,j}} [\exp[E(x_{2i-1,j}, u_{2i-1,j}, x_{2i-2,j}, u_{2i-2,j}) \\ &\qquad + E(x_{2i-1,j}, u_{2i-1,j}, x_{2i-1,j-1}, u_{2i-1,j-1})] \\ &\qquad \cdot \frac{1}{Z_{xu}^{2i,n-j+1}} \\ &\qquad \times \exp[E(x_{2i,n-j+1}, u_{2i,n-j+1}, \\ &\qquad\quad x_{2i-1,n-j+1}, u_{2i-1,n-j+1}) \\ &\qquad\quad + E(x_{2i,n-j+1}, u_{2i,n-j+1}, \\ &\qquad\quad\quad x_{2i,n-j+2}, u_{2i,n-j+2})] \Bigg\}. \end{aligned} \tag{11}$$

Then, the high-order DQNMF potential can be obtained in the log-domain for stability

$$E^{\mathrm{DQNMF}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}}) = \log[p(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}})]. \tag{12}$$

The high-order DQNMF potential considers the different label structure patterns regulated by the auxiliary field and captures the multidimensional label interactions. That is to say, it can predict the probability of the hidden random variable at (i', j') taking some label based on its spatial neighbors.

2) POE Potential Regulated by the Auxiliary Field: Second, in the modeling of the multidimensional label interactions, we focus on the regions' labeling consistency for pixels within the weak-structured region, and then a POE potential for the pixels within the weak-structured region is incorporated into the high-order energy function to enforce the regions' labeling consistency of HOTCRF-PCANet.

Our contribution is that we use the auxiliary field to classify the label structure patterns, and only if the centered site (i', j') belongs to the weak-structured region, i.e., $u_{i',j'} = g_1$, we incorporate the POE potential $E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}) \cdot \delta(u_{i',j'}, g_1)$ into the high-order energy function to enforce the label consistency cost. In fact, in the weak-structured region, the highly dependence of spatial neighbors will favor all the sites on low-order cliques (consisting of single and the pairwise cliques) within $S_{\mathrm{HO}_{(i',j')}}$ taking the same label configurations. And the baseline is the low-order label configurations centered at site (i', j'), that are $x_{i',j'} = e_l$, $(x_{i',j'}, x_{i',j'-1})_{\{(i',j'),(i',j'-1)\}\in PC_2(1)} = (e_l, e_{t1})$, $(x_{i',j'}, x_{i',j'+1})_{\{(i',j'),(i',j'+1)\}\in PC_2(2)} = (e_l, e_{t2})$, $(x_{i',j'}, x_{i'-1,j'})_{\{(i',j'),(i'-1,j')\}\in PC_2(3)} = (e_l, e_{t3})$ and $(x_{i',j'}, x_{i'+1,j'})_{\{(i',j'),(i'+1,j')\}\in PC_2(4)} = (e_l, e_{t4})$, where $e_l \in \Omega$, $e_{ta} \in \Omega, 1 \leq a \leq 4$, $PC_2 = \{PC_2(k)\}_{1\leq k\leq 4}$ denote the pairwise cliques within 4-neighborhood. Then, the POE potential $E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}) \cdot \delta(u_{i',j'}, g_1)$ is expressed as

$$\begin{aligned} & E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}) \cdot \delta(u_{i',j'}, g_1) \\ &\qquad = \log[f(e_l) \cdot \prod_{a=1}^{4} f(e_l, e_{ta})] \cdot \delta(u_{i',j'}, g_1) \end{aligned} \tag{13}$$

$$f(e_l) = n_{e_l}/M_{SC_1}, \quad f(e_l, e_{ta}) = n_{(e_l,e_{ta})}/M_{PC_2(k)} \tag{14}$$

where n_{e_l} denotes the number of samples in $\mathbf{x}_{\mathbf{HO}_{(i',j')}}$ taking each label e_l, and $n_{(e_l,e_{ta})}$ denotes the number of pairwise samples in $\mathbf{x}_{\mathbf{HO}_{(i',j')}}$ taking the labels (e_l, e_{ta}), M_{SC_1} and $M_{PC_2(k)}$ are the total numbers of the samples on single clique and the pairwise samples on pairwise clique $PC_2(k)$ within $\mathbf{x}_{\mathbf{HO}_{(i',j')}}$, respectively, $f(e_l)$ measures the probability of the hidden random variables in $\mathbf{x}_{\mathbf{HO}(i',j')}$ taking the label e_l, and $f(e_l, e_{ta})$ measures the probability of the hidden random variables taking the labels (e_l, e_{ta}). $u_{i',j'} = g_1$ represents that (i', j') is within a weak-structured region, $\delta(u_{i',j'}, g_1)$ verifies $\delta(u_{i',j'}, g_1) = 1$ for $u_{i',j'} = g_1$, and $\delta(u_{i',j'}, g_1) = 0$ for $u_{i',j'} \neq g_1$. Then, $E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}})$ favors all the sites on low-order cliques taking the same label configurations given the high-order neighborhood system $S_{\mathrm{HO}_{(i',j')}}$. The procedure of the construction of the POE potential is listed in Table I.

TABLE I
CONSTRUCTION OF POE POTENTIAL

1. Set the size of the overlapping high-order neighbourhood as 7×7.
2. If the centered site (i', j') of the overlapping high-order neighbourhood $S_{\mathrm{HO}_{(i',j')}}$ belongs to the weak-structured region, construct the POE potential as follows.
3. Set the baseline. The baseline is the low-order label configurations centered at site (i', j'), that are
   $x_{i',j'} = e_l$,
   $(x_{i',j'}, x_{i',j'-1})_{\{(i',j'),(i',j'-1)\}\in C_2(1)} = (e_l, e_{t1})$,
   $(x_{i',j'}, x_{i',j'+1})_{\{(i',j'),(i',j'+1)\}\in C_2(2)} = (e_l, e_{t2})$,
   $(x_{i',j'}, x_{i'-1,j'})_{\{(i',j'),(i'-1,j')\}\in C_2(3)} = (e_l, e_{t3})$, and
   $(x_{i',j'}, x_{i'+1,j'})_{\{(i',j'),(i'+1,j')\}\in C_2(4)} = (e_l, e_{t4})$.
4. Compute the frequency of each label configuration above in $\mathbf{x}_{\mathbf{HO}(i',j')}$ by the normalized joint histogram (NJH) method [15], and thus obtain the corresponding probabilities based on (14): $f(e_l)$ and $f(e_l, e_{ta}), 1 \leq a \leq 4$.
5. Construct the POE potential. Take the probability of each label configuration as an expert to model the high-order label dependency within $\mathbf{x}_{\mathbf{HO}(i',j')}$ and then we have the POE potential that provides the label consistency cost, as given by (13).

B. Unary Potential Based on Unsupervised PCANet

In this section, we modify PCANet to fit unsupervised segmentation, named as UPCANet. UPCANet is able to automatically extract hierarchical features from SAR images, rather than handcrafted features. Then, we integrate UPCANet into the CRF framework for unsupervised SAR image segmentation, and construct an UPCANet-based unary potential to predict the class probability using deep features.

First, following the way in [27], we illustrate how UPCANet extracts features for unsupervised image segmentation. In the

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first stage, for each class e_l, UPCANet randomly selects N fake training samples from the current segmentation result. Around each fake sample $h, 1 \le h \le N$, we have a $np \times np$ subimage Υ_h. We will use a 2-D convolutional filter whose size is $k \times k$ to extract deep feature maps from each subimage. First, we collect overlapping patches of size $k \times k$ from Υ_h, and the number of the collected overlapping patches is $Q = (np - (k-1))^2$. Then, we subtract the patch mean from each patch, arrange each patch into a vector, and thus obtain $C_h = [c_{h,1}, \ldots, c_{h,Q}] \in \mathbb{R}^{kk \times Q}$, where $c_{h,z} \in \mathbb{R}^{kk}, 1 \le z \le Q$ is the mean-removed vector (1-dimension). We construct the same matrix for all the fake samples, and combine them into a matrix $C = [C_1, C_2, \ldots, C_N] \in \mathbb{R}^{kk \times QN}$.

Then, the eigenvectors of CC^T are computed, and we take H_1 principal eigenvectors to form the PCA filter that is formulated as

$$W_d^1 = \text{mat}(q_d(CC^T)) \in \mathbb{R}^{k \times k}, \quad 1 \le d \le H_1 \tag{15}$$

where $q_d(CC^T)$ denotes the dth principal eigenvector of CC^T, mat($\hbar$) is a function that maps $\hbar \in \mathbb{R}^{kk}$ to a matrix $W \in \mathbb{R}^{k \times k}$. More details can be found in [27].

Now we can extract the first-stage features using the principal eigenvectors

$$\Upsilon_h^d = \Upsilon_h * W_d^1, \quad 1 \le h \le N \tag{16}$$

where $*$ represents a 2-D convolution operation. The boundary of Υ_h is padded before convolution to make Υ_h^d have the same size with Υ_h.

Then, we obtain $N \times H_1$ outputs from the first stage. In the second stage, for each $\Upsilon_h^d, 1 \le d \le H_1$, we collect all the patches of Υ_h^d as in the first stage, and form $J_h^d = [j_{h,d,1}, \ldots, j_{h,d,Q}] \in \mathbb{R}^{kk \times Q}$, where $j_{h,d,z} \in \mathbb{R}^{kk}, 1 \le z \le Q$ is the vectorized mean-removed patch in Υ_h^d. By concatenating $J_h^d, 1 \le d \le H_1$, we have $J^d = [J_1^d, \ldots, J_N^d] \in \mathbb{R}^{kk \times NQ}$. By combining the H_1 filter outputs, the inputs for the second layer of UPCANet are obtained

$$J = [J^1, \ldots, J^{H_1}] \in \mathbb{R}^{kk \times H_1 NQ}. \tag{17}$$

As the case in the first stage, the first H_2 principal eigenvectors of JJ^T are obtained to form the PCA filter in the second layer $W_b^2, 1 \le b \le H_2$

$$W_b^2 = \text{mat}(q_d(JJ^T)) \in \mathbb{R}^{k \times k}, 1 \le b \le H_2. \tag{18}$$

By convolving $W_b^2, 1 \le b \le H_2$ with each output of the first layer $\Upsilon_h^d, 1 \le d \le H_1, 1 \le h \le N$, the second-stage features can be obtained

$$\text{O}_h^{d,b} = W_b^2 * \Upsilon_h^d \in \mathbb{R}^{k \times k}, \quad 1 \le d \le H_1, 1 \le b \le H_2 \tag{19}$$

Then for the subimage Υ_h, the deep features obtained by UPCANet are expressed as

$$\text{O}_h = \left\{\text{O}_h^{1,1}, \text{O}_h^{1,2}, \ldots, \text{O}_h^{1,H_2}, \ldots, \text{O}_h^{H_1,1}, \text{O}_h^{H_1,2}, \ldots, \text{O}_h^{H_1,H_2}\right\}. \tag{20}$$

Now we discuss how to construct the UPCANet-based unary potential for segmentation. In the training stage, UPCANet randomly selects N fake training samples for each class e_l, from current segmentation result. Then the deep feature maps of the fake training samples are extracted by UPCANet as discussed above. More specifically, for each class e_l, we construct the two-stage PCA filters $W_{d,l}^1, 1 \le d \le H_1$ and $W_{b,l}^2, 1 \le b \le H_2$ by using (15) and (18), respectively, and extract the baseline deep feature maps (BDFM) $\text{O}_{h,l}^{\text{BDFM}}, 1 \le l \le L, 1 \le h \le N$ by using (20).

Then, for the segmented image, around each pixel, we extract an overlapping subimage with the size being $np \times np$. Then the two-stage PCA filters $W_{d,l}^1, 1 \le d \le H_1$ and $W_{b,l}^2, 1 \le b \le H_2$ are utilized to extract deep feature maps. Assuming that the subimage is centered at site (i', j'), the extracted testing deep feature maps (TDFM) at site (i', j') are denoted as $\text{O}_{(i',j'),l}^{\text{TDFM}}, 1 \le l \le L$.

Given the deep feature maps by UPCANet, the dissimilarity measure between the TDFM and BDFM is expressed as

$$D_{(i',j'),l} = \sum_{h=1}^{N} \left\|\text{O}_{(i',j'),l}^{\text{TDFM}} - \text{O}_{h,l}^{\text{BDFM}}\right\|^2, \quad 1 \le l \le L \tag{21}$$

and the unary potential at site (i', j') can be expressed as a function of the dissimilarity measure

$$A^{\text{UPCANet}}(x_{i',j'} = e_l | y) = \exp(-\tau D_{(i',j'),l}), \quad 1 \le l \le L \tag{22}$$

where τ is a weighting factor that was fixed empirically to the same value in all experiments. The procedure of the construction of the UPCANet-based unary potential is listed in Table II.

TABLE II
CONSTRUCTION OF UPCANET-BASED UNARY POTENTIAL

1. Extract N fake samples for each class from the current segmentation, and around each fake sample, construct a $np \times np$ subimage Υ_h, $h, 1 \le h \le N$.
2. For each class e_l, construct the baseline deep feature maps:
   - 2.1. Use the corresponding fake samples to learn the PCA filters $W_{d,l}^1, 1 \le d \le H_1$ and $W_{b,l}^2, 1 \le b \le H_2$ by using (15) and (18);
   - 2.2. Convolve the filters with subimage Υ_h, $h, 1 \le h \le N$, and take the obtained feature maps $\text{O}_{h,l}^{BDFM}, 1 \le l \le L, 1 \le h \le N$ as the BDFM of class e_l;
3. For the pixel at site (i', j') for segmenting, construct the UPCANet-based unary potential associated with class e_l:
   - 3.1. Use the learned PCA filters $W_{d,l}^1, 1 \le d \le H_1$ and $W_{b,l}^2, 1 \le b \le H_2$ to obtain the TDFM, $\text{O}_{(i',j'),l}^{TDFM}, 1 \le l \le L$
   - 3.2. Measure the dissimilarity between the obtained TDFM and BDFM by using (21)
   - 3.3. Express the unary potential associated with class e_l as a function of the dissimilarity measure by using (22).

C. Likelihood

Likelihood reflects the physical electromagnetic scattering mechanisms and is important to SAR image segmentation [41]–[43]. Assuming that the random variables $Y = (Y_{i',j'})_{(i',j') \in S}$ are independent conditional to $(X, U) = (X_{i',j'}, U_{i',j'})_{(i',j') \in S}$ classically, and the distribution of each

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TABLE III
SEGMENTATION PROCEDURE OF HOTCRF-PCANET

1. Initialize the segmentation of the label and auxiliary fields using the CRF model [15] and NAGK respectively.
2. Construct the local high-order energy function by using (6).
3. Randomly select the fake samples from the current segmentation, and construct the UPCANet-based unary potential by using (22).
4. Construct the data term $p(y_{i',j'} \mid x_{i',j'})$ using GGamma distribution to model the SAR scattering statistics.
5. Model the posterior distribution of HOTCRF-PCANet by using (5).
6. MPM segmentation is utilized for labeling.
7. Estimate the model parameters of HOTCRF-PCANet, $\alpha = \{\alpha_H^1, \alpha_{mH}^2, \alpha_V^1, \alpha_{mV}^2\}, 1 \le m \le M$ and $\kappa_l = \{\vartheta_l, \beta_l, \lambda_l\}$, using ICE method as discussed in Subsection V-B.
8. If maximum number of MPM iteration is not completed, go to step 2.

$Y_{i',j'}$ conditional to (X, U) is equal to its distribution conditional to $X_{i',j'}$, we have the likelihood

$$p(y|x,u) = \prod_{(i',j')\in S} p(y_{i',j'}|x_{i',j'}). \tag{23}$$

GGamma distribution is utilized to model the SAR scattering statistics [41], and then the likelihood term of site (i', j') can be expressed as

$$p(y_{i',j'}|x_{i',j'} = e_l) = \frac{\beta_l (y_{i',j'})^{\beta_l\lambda_l - 1}}{\vartheta_l^{\beta_l\lambda_l}\Gamma(\lambda_l)} \exp\left[-\left(\frac{y_{i',j'}}{\vartheta_l}\right)^{\beta_l}\right] \tag{24}$$

where $\kappa_l = \{\vartheta_l, \beta_l, \lambda_l\}$ is the GGamma distribution parameters corresponding to class l, and are estimated by the method of logarithmic cumulants (MoLC) [43] in the segmentation.

D. Inference

In the inference, we use the maximization of posterior marginals (MPM) estimator which minimizes the expected number of the mislabeled sites and is more reasonable for the image segmentation [44]. First, we use Gibbs sampler to alternately generate the discrete-time Markov chains $x' = \{x'(1), \ldots, x'(N_{mc})\}$ and $u' = \{u'(1), \ldots, u'(N_{mc})\}$ according to the posterior distribution. Then the marginal posterior probability $p(x_{i',j'} = e_l|u, y)$ is approximated by $p(x_{i',j'} = e_l|u, y) \approx (1/N_{mc}) \sum_{b=1}^{N_{mc}} z_{l,(i',j')}(b)$, where $z_{l,(i',j')}(b) = 1$ for $x'_{i'j'}(b) = e_l$, $z_{l,(i',j')}(b) = 0$ otherwise, and N_{mc} is the number of visits to pixel (i', j') by Gibbs sampler and is set as 8 empirically, and $p(u_{i',j'} = g_m|x, y)$ is obtained in the same way. Then the MPM inference can be stated as

$$\begin{aligned} x_{i',j'} &= \arg\max_{e_l\in\Omega} p(x_{i',j'} = e_l|u, y) \\ u_{i',j'} &= \arg\max_{g_m\in\Lambda} p(u_{i',j'} = g_m|x, y). \end{aligned} \tag{25}$$

The segmentation procedure of HOTCRF-PCANet is listed in Table III.

V. PARAMETER SETTING AND ESTIMATION

A. Parameter Setting of HOTCRF-PCANet

HOTCRF-PCANet needs to set the parameters of UPCANet, including subimage size, patch size, the number of principal eigenvectors, and the number of fake sample. The parameters of UPCANet are set as follows empirically:

1) The size of the subimage is set as 7×7, and then $np = 7$;
2) The patch size is set as 7×7.
3) At the first stage, the first 8 principal eigenvectors are empirically used to form the PCA filter in the first layer, and then $H_1 = 8$. At this stage, the size of PCA filter is $49 \times H_1 \times L$, where L is the number of classes.
4) At the second stage, the first 3 principal eigenvectors are empirically extracted, and then $H_2 = 3$. At this stage, the size of PCA filters is $49 \times H_2 \times L$, where L is the number of classes.
5) N is set as 100. That is to say 100 fake samples are used for each class to train the UPCANet.
6) In addition, in the construction of the high-order DQNMF potential and POE potential, a 7×7 high-order neighborhood system is taken into account.

B. Parameter Estimation of HOTCRF-PCANet

In the proposed HOTCRF-PCANet, we have two sets of parameters to be estimated. One is the set of interaction parameters $\alpha = \{\alpha_H^1, \alpha_V^1, \alpha_{mH}^2, \alpha_{mV}^2\}, 1 \le m \le M$ and the other is the set of GGamma distribution parameters $\eta = \{\kappa_l\}, 1 \le l \le L$, where M and L denote the numbers of stationarity and classes.

Let $\tilde{\theta}(X, U, Y)$ be an estimator of the model parameters $[\alpha, \eta]$. Then we have to approach $\tilde{\theta}(X, U, Y)$ by a function of Y. And the best approximation is the conditional expectation as far as the mean square error is concerned. Consequently, we utilize the scheme of iterative conditional estimation (ICE) [45] to estimate the model parameters. ICE is more general than expectation maximization (EM) and is often easier to perform, and it has been successfully applied in random fields model [31], [46], [47]. The ICE is defined as the following iteration procedure: 1) Initialization θ^0; 2) $\theta^{T+1} = E[\tilde{\theta}(X, U, Y)|Y = y, \theta^T]$, where T denotes the ICE iteration, and $E[\tilde{\theta}(X, U, Y)|Y = y, \theta^T]$ is the expectation of $\tilde{\theta}(X, U, Y)$ according to the distribution of (X, U) conditioned on Y based on θ^T, that is the posterior distribution.

However, the calculation of the conditional expectation, such as $E[\tilde{\alpha}_H^1(X, U, Y)|Y = y, \theta^T]$, is not possible in practice. Fortunately, sampling of (X, U) according to the distribution conditioned on Y is possible, so we can approach it using the empirical mean by virtue of the law of large numbers. That is to say, using Gibbs sampler, we alternately sample X and U according to the posterior distribution, and obtain A realizations $\{(\tilde{x}(1), \tilde{u}(1)), (\tilde{x}(2), \tilde{u}(2)), \ldots, (\tilde{x}(A), \tilde{u}(A))\}$ of (X, U). Then, we can estimate the parameter θ^{T+1} by $\theta^{T+1} = (1/A)(\tilde{\theta}[\tilde{x}(1), \tilde{u}(1)] + \tilde{\theta}[\tilde{x}(2), \tilde{u}(2)] + \cdots + \tilde{\theta}[\tilde{x}(A), \tilde{u}(A)])$.

Then we present how to compute $\tilde{\theta}(\tilde{x}(a), \tilde{u}(a), y)$ for each obtained $(\tilde{x}(a), \tilde{u}(a)), 1 \le a \le A$. Let $Y = (Y_s)_{s\in S}$ stand for the initial model parameters. Then let θ^T stand for the estimation of model parameters θ obtained at the Tth iteration. Then the estimation procedure of the model parameters is as follows:

1) Using Gibbs sampler, we alternately sample X and U according to the posterior distribution, and obtain A realizations $\{(\tilde{x}(1), \tilde{u}(1)), (\tilde{x}(2), \tilde{u}(2)), \ldots, (\tilde{x}(A), \tilde{u}(A))\}$

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of (X, U) corresponding to θ^T and Y. Note that Gibbs sampler is used independently A times.

2) For each obtained set $(\tilde{x}(a), \tilde{u}(a)), 1 \le a \le A$, use Gibbs sampler to alternately simulate B new realizations, $(x'(\tilde{b}), u'(\tilde{b})), 1 \le \tilde{b} \le B$, of X and U according to $(1/Z^x_{HO}) \exp\{\sum_{(i',j')\in S}[E^{\mathrm{DQNMF}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}}) + E^{\mathrm{POE}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}})]\}$, and $(1/Z^u_{HO}) \exp\{\sum_{(i',j')\in S} E^{\mathrm{POE}}(\mathbf{u}_{\mathbf{HO}_{(i',j')}})\}$, respectively, where Z^x_{HO} and Z^u_{HO} are the partition functions. Let $(x'(0), u'(0)) = (\tilde{x}(a), \tilde{u}(a))$ and estimate the model parameter $\{\alpha^1_H, \alpha^2_{mH}\}, 1 \le m \le M$ based on stochastic gradient [47] as follows:

$$\alpha^1_H(\tilde{b}) = \alpha^1_H(\tilde{b}-1) + \frac{\mathrm{F}}{\tilde{b}}\left[\nabla_{\alpha^1_H} E^{\mathrm{HO}}(x'(\tilde{b}), u'(\tilde{b})) - \nabla_{\alpha^1_H} E^{\mathrm{HO}}(x'(0), u'(0))\right]$$
$$\alpha^2_{mH}(\tilde{b}) = \alpha^2_{mH}(\tilde{b}-1) + \frac{\mathrm{F}}{\tilde{b}}\left[\nabla_{\alpha^2_{mH}} E^{\mathrm{HO}}(x'(\tilde{b}), u'(\tilde{b})) - \nabla_{\alpha^2_{mH}} E^{\mathrm{HO}}(x'(0), u'(0))\right] \quad (26)$$

where $(x'(\tilde{b}), u'(\tilde{b})), 1 \le \tilde{b} \le B$ is the realization of (X, U) using the current model parameters, and F is a constant that is set as 1 in the experiment. We have the same scheme for the estimation of $\{\alpha^1_V, \alpha^2_{mV}\}, 1 \le m \le M$. In practice, A is set as 2, and B is set as 1 empirically in the experiment. Then we can obtain the estimated interaction parameters $\tilde{\alpha}[\tilde{x}(a), \tilde{u}(a)] = \{\alpha^1_H(a), \alpha^1_V(a), \alpha^2_{mH}(a), \alpha^2_{mV}(a)\}, 1 \le m \le M$.

3) MoLC [43] is used to estimate the GGamma parameters and the estimated parameters are denoted as $\tilde{\kappa}_l(a) = \{\tilde{\vartheta}_l(a), \tilde{\beta}_l(a), \tilde{\lambda}_l(a)\}$. Then we have the estimation of model parameters $\tilde{\theta}[(\tilde{x}(a), \tilde{u}(a))]$.

4) Let $\tilde{\theta}[(\tilde{x}(1), \tilde{u}(1))], \tilde{\theta}[(\tilde{x}(2), \tilde{u}(2))], \ldots, \tilde{\theta}[(\tilde{x}(A), \tilde{u}(A))]$ be the estimated values. θ^{T+1} can be estimated by

$$\theta^{T+1} = \frac{1}{A}\{\tilde{\theta}[(\tilde{x}(1), \tilde{u}(1))] + \tilde{\theta}[(\tilde{x}(2), \tilde{u}(2))] + \cdots + \tilde{\theta}[(\tilde{x}(A), \tilde{u}(A))]\}. \quad (27)$$

If the sequence of the parameters θ^T becomes steady, the iterations are finished.

VI. Experimental Results and Discussion

A. Data Sets Description

Simulated images and real SAR images are segmented in the experiment to illustrate the effectiveness and popularity of the proposed HOTCRF-PCANet. First, we have a simulated image shown in Fig. 5(a). There are two classes in this image, and this simulated image can be taken as the ground truth. Then, we add multiplicative noises with different noise variances to this simulated image. In this way, we have four simulated noisy images, whose noise variances are 0.167, 0.25, 0.5, and 1, respectively. The four simulated noisy images are shown in Fig. 5(b)–(e), respectively. And the corresponding effective numbers of looks (ENLs) of this set of simulated noisy images are 6-look, 4-look, 2-look, and 1-look, respectively. To evaluate the segmentation performance for the simulated images, overall accuracy and Kappa parameter are utilized [48]. The second set of simulated image is obtained as follows. First, we extract the samples from the panes with three different colors in experimental airborne SAR (ESAR) image, as shown in Fig. 6(a). Each color denotes a class in fact. Then, the samples extracted above are manually filled into the ground truth image which is shown in Fig. 6(b). The obtained simulated image is shown in Fig. 6(c). The prior information of real SAR images is shown in Table IV.

TABLE IV
Prior Information of Real SAR Images

Image	Size	Class	Band	Image source
Real SAR 1	480×480	4	X	Domestic airborne SAR data
Real SAR 2	440×440	4	Ku	Sandia National Laboratory
Real SAR 3	400×400	5	X	TerraSAR
Real SAR 4	512×512	5	X	Domestic airborne SAR data

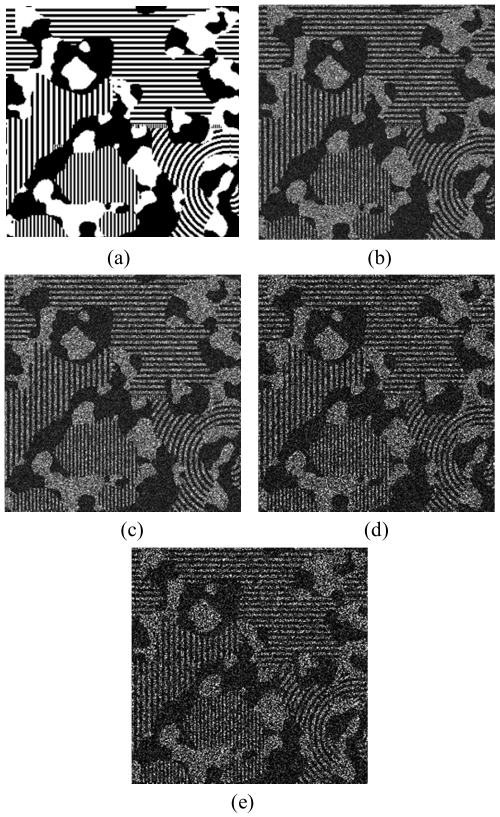

Fig. 5. Simulated images with the ENLs being 6, 4, 2, and 1, respectively. (a) Ground truth. (b) 6-look. (c) 4-look. (d) 2-look. (e) 1-look.

B. Reference Models

In the experiment, the following models are taken as the reference model:

1) Convolutional neural networks (CNN) is a popular form of deep learning networks that is able to learn hierarchical contextual features from the image by the convolutional kernels [21]. CNN has achieved very promising results in remote sensing image segmentation recently. Consequently, we take CNN as a reference method in

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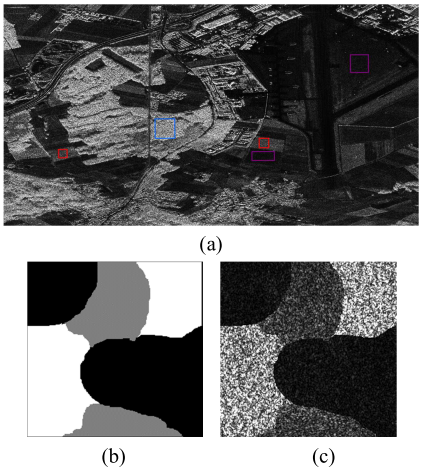

Fig. 6. Simulated image obtained from ESAR image. (a) ESAR image. (b) Ground truth. (c) Simulated image.

this article. The basic structure of the CNN in our experiment is a two-layer convolution. In this CNN, the padding mode is SAME, and we perform a batch normalization (BN) operation on the result of the linear convolution, and then pass through a Relu nonlinear layer. The kernel size in the two-layer convolution is set as 3. The number of filters in the first layer and the second one are 32 and 64, respectively. CNN extracts features from 7×7 overlapping window centered at this pixel and thus predicts the class. The number of parameters for 2, 4, and 5 classes are 4000, 7136, and 8704, respectively. In the training of CNN, 200 training samples are used for each class, and the batch size and number of epochs are set 100 and 200, respectively. Then, for the images with 2, 4, and 5 classes, we need 800, 1600, and 2000 steps, respectively, to learn the parameters in the training.

2) Inception-ResNet [24]. Inception-ResNet is a typical deep convolutional network that has achieved very good performance in segmentation, and more details about Inception-ResNet can be found in [24]. Inception-ResNet extracts features from 7×7 overlapping window centered at this pixel and thus predicts the class. The number of parameters for 2, 4, and 5 classes are 16992, 23264, and 35808, respectively. In the training of Inception-ResNet, 400 training samples are used for each class, and the batch size and number of epochs are set 100 and 100, respectively. Then, for the image with 2, 4, and 5 classes, we need 800, 1600, and 2000 steps, respectively, to learn the parameters in the training.
3) CRF-CNN [25]. CRF-CNN integrates convolution network with CRF model. Thus, it uses the deep features to predict the label and incorporates the spatial correlation into the segmentation process. CRF-CNN has proven its superiority in natural image segmentation and is also taken as the reference model. In CRF-CNN, the convolutional network is pretrained as discussed above in CNN, and is utilized to construct the unary potential.
4) A HN-TMF model [16]. In HN-TMF, the local energy functions are defined either in a homogeneous or heterogeneous way based on the auxiliary field. Then, HN-TMF model captures the high-order label dependences and the prior modeling is effectively enhanced.

C. Validity of High-Order DQNMF Potential

The high-order DQNMF potential $E^{\mathrm{DQNMF}}(\mathbf{x}_{\mathbf{HO}_{(i',j')}}, \mathbf{u}_{\mathbf{HO}_{(i',j')}})$ is proposed to predict the probability of the hidden random variable at (i', j') taking some label based on its high-order spatial neighbors. In this section, we test the validity of the proposed high-order DQNMF potential. The simulated image in this experiment is shown in Fig. 7(a). It is corrupted by multiplicative noise and the equivalent number of looks (ENL) is 4. The ground truth is given by Fig. 7(b).

From Region 1 and Region 2, we randomly select 50 samples that belong to class 1, respectively. Given a 7×7 high-order neighborhood system, we compute the high-order DQNMF potentials w.r.t these samples. As a comparison, the pairwise potentials on a 4-neighboring system are also measured on the sample sites. The pairwise potential values w.r.t these samples are given in Fig. 7(c) and (e), while the high-order DQNMF potential values w.r.t these samples are given in Fig. 7(d) and (f). In Fig. 7(c)–(f), each sample site has two potential values corresponding to the two candidate classes in $\Omega = \{1, 2\}$, and we use blue and pink to distinguish the potential values of the two classes.

According to the potential values shown in Fig. 7, the proposed high-order DQNMF potential is able to predict the class of the sample more accurately. That is to say, the high-order DQNMF potential values associated with class 1 are almost great larger than those associated with class 2, as shown in Fig. 7(d) and (f). In comparison with the high-order DQNMF potential, the pairwise potential gives weak discrimination relatively for these samples, as shown in Fig. 7(c) and (e). In addition, for some samples within the blue panes in Fig. 7(c) and (e), inaccurate potential values are achieved by the pairwise potential, which means that the potential values associated with class 1 are smaller than those associated with class 2.

And fortunately, these samples can be discriminated successfully by the proposed high-order DQNMF potential. The reason can be attributed to its enhanced ability of modeling multidimensional label configuration. Because the potential value is obtained according to the label configuration, we can say that compared to the pairwise potential, the high-order DQNMF potential can capture these two intense label structures more accurately and thus obviously provides more powerful discrimination in the labeling.

D. Validity of UPCANet for SAR Image Segmentation

UPCANet, that is a simple but highly competitive deep learning network, is utilized to deal with image segmentation. The UPCANet-based unary potential can be constructed as discussed in Section IV-B. In this section, we test the validity of the proposed UPCANet-based unary potential. The parameters

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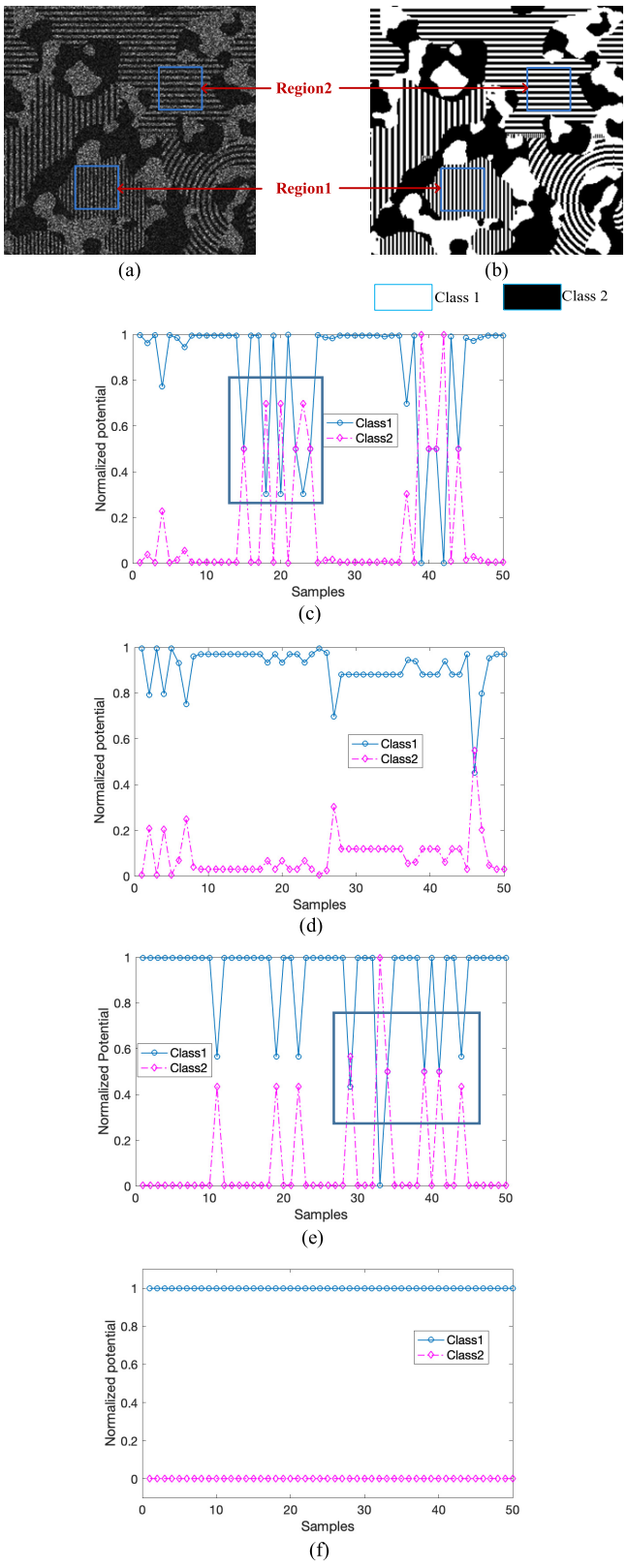

Fig. 7. Comparison of the high-order DQNMF potential and the pairwise potential in the modeling of label configuration. (a) Simulated image (4-look). (b) Ground truth. (c) Pairwise potentials of samples belonging to class1 in region1. (d) High-order DQNMF potentials of samples belonging to class1 in region1. (e) Pairwise potentials of samples belonging to class1 in region2. (f) High-order DQNMF potentials of samples belonging to class1 in region2.

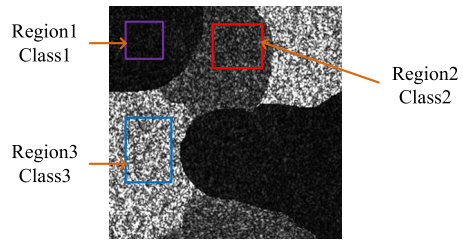

Fig. 8. Test regions of the simulated image from ESAR image.

of UPCANet are given in Section V-A. As a comparison, CNN is taken as the reference model and the output of soft-max is taken as the CNN-based unary potential. In this experiment, the simulated image from ESAR image is utilized, as shown in Fig. 8. For each class, we extract 100 fake samples randomly to train the UPCANet, and 200 samples for each class are extract randomly to train CNN, whose parameters are given in Section VI-B. Then, UPCANet-based and CNN-based unary potentials can be obtained.

To test the validity of the unary potentials, we randomly extract 100 samples from region 1, 2, and 3, respectively, and each set of 100 samples from region 1, 2, and 3 belongs to a certain class in fact, as shown in Fig. 8. The unary potentials of these samples are given in Fig. 9. Each sample site has three potential values corresponding to the three candidate classes in $\Omega = \{1, 2, 3\}$. We use blue, orange, and pink to distinguish the potential values of the three classes. The UPCANet-based unary potentials are shown in Fig. 9(a), (c), and (e), while the CNN-based ones are presented in Fig. 9(b), (d), and (f). As shown in Fig. 9(a), (c), and (e), the UPCANet-based unary potentials have reliable discrimination ability in the labeling. That is to say, UPCANet achieves reliable unary potentials to label these samples effectively. For instance, as shown in Fig. 9(e), the potential values associated with class 3 are almost great larger than those associated with class 2 and class 1. Overall, the discrimination ability of UPCANet is competitive to CNN to some extent. That is to say, UPCANet has a feasible representation-learning ability for image segmentation, and can offer a discriminating unary potential in our proposed HOTCRF-PCANet.

E. Experiment and Discussion on Simulated Images

In this section, we use the simulated images corrupted by the multiplicative noises, with ENLs being 6, 4, 2, and 1, shown in Fig. 5(b)–(e). CNN [21], Inception-ResNet [24], CRF-CNN [25], and HN-TMF [16] are taken as the referenced methods. 200 and 400 training samples are extracted from the ground truth randomly to train the CNN and Inception-ResNet, respectively. The implementation details of CNN and Inception-ResNet are discussed in Section V-B.

The segmentation results of the reference models and HOTCRF-PCANet are shown in Figs. 10–13(a)–(e). For this set of simulated images, the auxiliary field is used to regulate three label structure patterns that are weak-structured, vertical-structured, and horizontal-structured patterns. The segmentation of the label structure pattern, denoted

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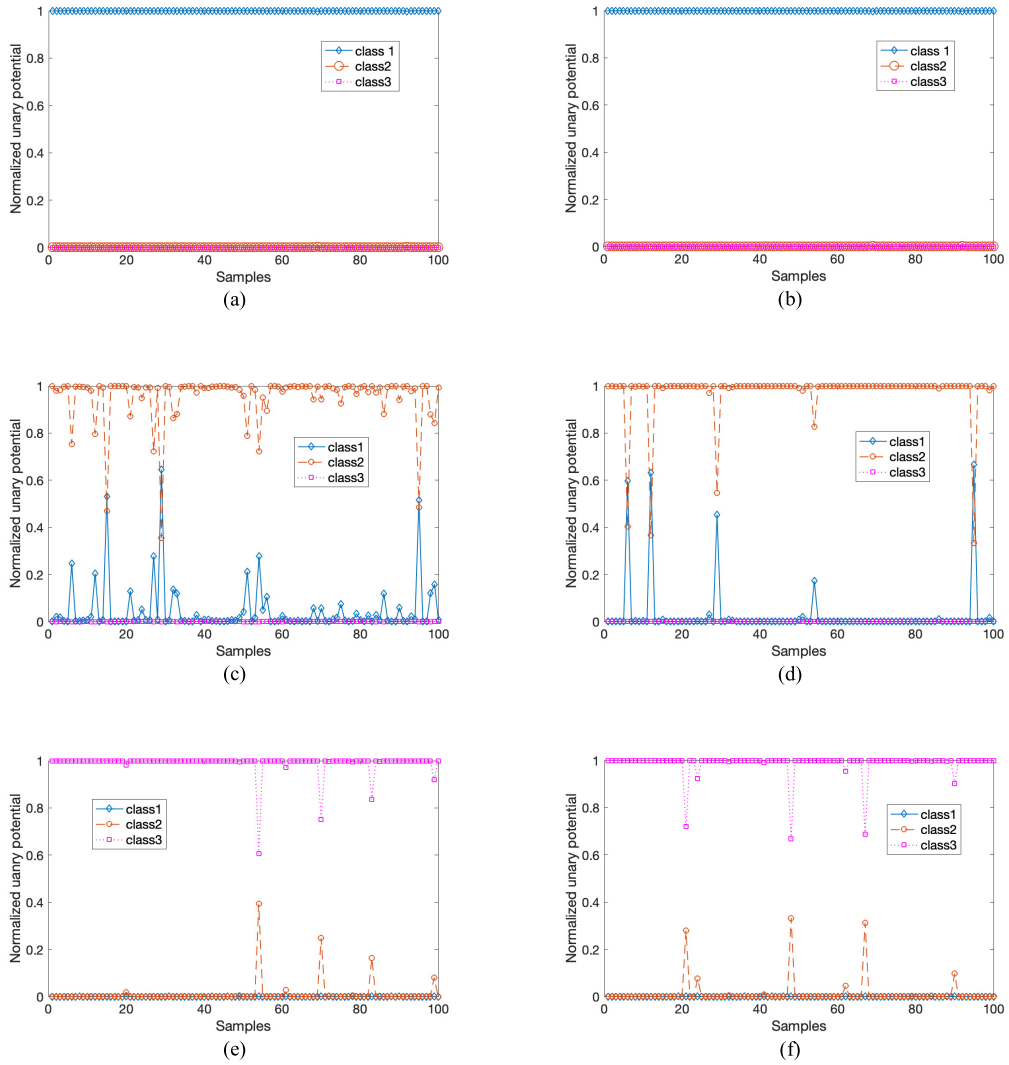

Fig. 9. UPCANet-based and CNN-based unary potentials. (a) UPCANet-based unary potentials of samples in region1. (b) CNN-based unary potentials of samples in region1. (c) UPCANet-based unary potentials of samples in region2. (d) CNN-based unary potentials of samples in region2. (e) UPCANet-based unary potentials of samples in region3. (f) CNN-based unary potentials of samples in region3.

by the auxiliary field U, is given in Figs. 10–13(f) in which black, gray, and white denote the weak-structured, vertical-structured, and horizontal-structured patterns, respectively. The estimated interaction parameters are listed in Table V. As listed in Table V, by using the auxiliary field to regulate the label structure patterns, HOTCRF-PCANet can effectively model the label interactions. For instance, in the vertical-structured region, the estimated α_{2V}^2 is larger than α_{2H}^2. That is to say, we pay more attention to the vertical label interaction in the segmentation. And in the horizontal-structured region, the estimated α_{3H}^2 is larger than α_{3V}^2, which means that the horizontal label interactions are emphasized. In this way, the nonstationary label structure information can be effectively captured by HOTCRF-PCANet.

As shown in Figs. 10–13(e), the proposed HOTCRF-PCANet can achieve better segmentation of the simulated images. HOTCRF-PCANet has the following advantages over the referenced methods. First, the region homogeneity is better. In addition, the segmentation of the intense-structured region is more accurate. That is to say, for the horizontal-structured and vertical-structured regions, HOTCRF-PCANet can achieve better segmentation accuracies and edge locations. The quantitative evaluation utilizing overall accuracy and Kappa parameter [48] is listed in Table VI, which also demonstrates the superiority of the proposed HOTCRF-PCANet. Furthermore, with the noise variance increasing, the superiority of HOTCRF-PCANet becomes more obvious, which demonstrates that HOTCRF-PCANet is more robust against multiplicative noises than the referenced models. For instance, with the noise variance increasing, the segmentation accuracies by CRF-CNN are 97.11%, 96.13%, 92.25%, and 87.39%, respectively, while those obtained by the proposed model being 98.49%, 97.33%, 94.65%, and 89.54%, respectively. The accuracy of CRF-CNN degrades a bit faster than that of HOTCRF-PCANet, which means that HOTCRF-PCANet is superior to CRF-CNN.

The improvement of HOTCRF-PCANet over the referenced models can be attributed as the following reason. Compared to the deep learning networks, CNN and Inception-ResNet, that use deep features to predict the label, HOTCRF-PCANet

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TABLE V
ESTIMATED INTERACTION PARAMETERS

	Intense-structured region (Horizontal structure)		Intense-structured region (Vertical structure)		Weak-structured region	
ENL	α_{3H}^2	α_{3V}^2	α_{2H}^2	α_{2V}^2	α_{1H}^2	α_{1V}^2
6-look	**0.9020**	0.1453	0.1352	**0.7314**	0.4439	0.5671
4-look	**1.0968**	0.1831	0.0616	**1.2186**	0.6033	0.8958
2-look	**1.0926**	0.2992	0.1539	**0.8572**	0.5830	0.5048
1-look	**1.2802**	0.2682	0.1987	**1.4482**	0.6294	0.9524

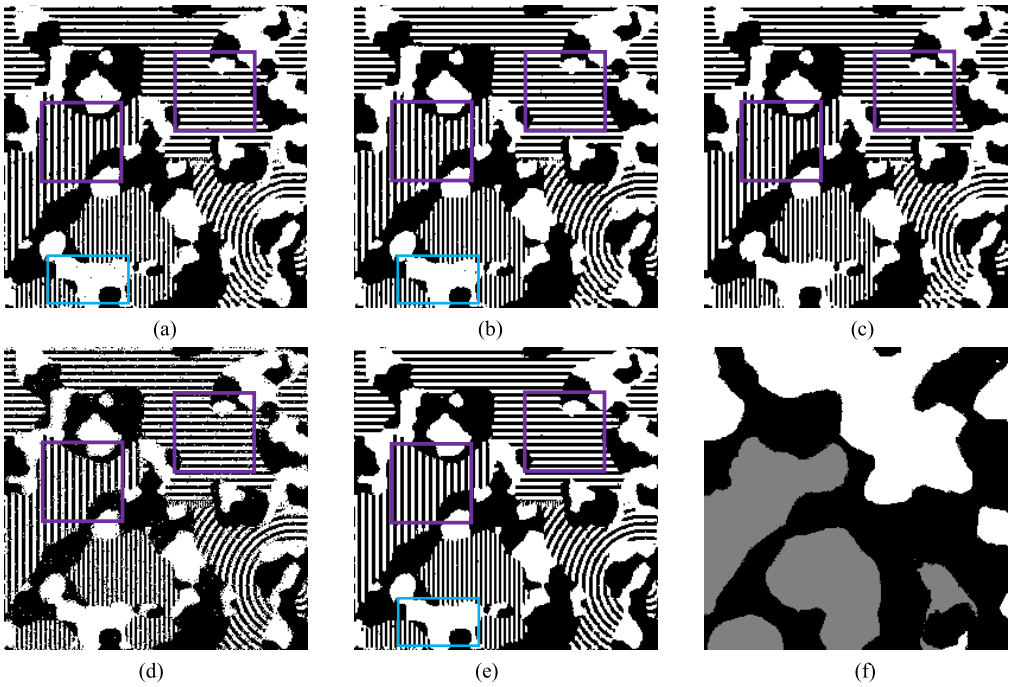

Fig. 10. Segmentation results of simulated image with ENLs being 6-look. (a) CNN. (b) Inception-ResNet. (c) CRF-CNN. (d) HN-TMF. (e) HOTCRF-PCANet. (f) Estimated U.

TABLE VI
QUANTITATIVE EVALUATION (ACCURACY/KAPPA) OF SEGMENTATION RESULTS FOR SIMULATED IMAGES

ENL	CNN	Inception-ResNet	CRF-CNN	HN-TMF	HOTCRF-PCANet
	Accuracy Kappa	Accuracy Kappa	Accuracy Kappa	Accuracy Kappa	Accuracy Kappa
6-look	96.69% 0.9391	97.34% 0.9468	97.11% 0.9420	96.91% 0.9380	98.49% 0.9696
4-look	95.31% 0.9060	96.38% 0.9275	96.13% 0.9225	94.90% 0.8975	97.33% 0.9463
2-look	90.84% 0.8160	92.76% 0.8544	92.25% 0.8442	85.06% 0.6962	94.65% 0.8927
1-look	85.45% 0.7077	85.53% 0.7086	87.39% 0.7466	71.84% 0.4205	89.54% 0.7550

models the nonstationary and high-order label interactions additionally. Concretely, the intense label structure is modeled by the proposed high-order DQNMF potential, and the label consistency for the pixels within weak-structured regions can be enforced effectively by the POE potential. Consequently, in the results shown in Figs. 10–13(a), 10–13(b), and 10–3(e), HOTCRF-PCANet preserves the structure information more accurately, as shown in the purple panes, and achieves better region homogeneities, as shown in the blue panes. HOTCRF-PCANet is also superior to CRF-CNN and HN-TMF. Although CRF-CNN takes the label interactions into account and achieves better results than CNN, the label interaction is limited within the pairwise cliques in four neighborhood system in fact. On the other hand, HN-TMF did not define a specific potential to capture the high-order label interactions. On comparison, the high-order DQNMF potential explicitly models the multidimensional label configuration within the high-order neighborhood, and thus captures the high-order label interactions effectively. Additionally, by using the deep features by PCANet, HOTCRF-PCANet predicts the class probability more accurately than HN-TMF. Thus, the segmentation of the intense-structured and the weak-structured regions by HOTCRF-PCANet is better, as shown in the blue and purple panes in Figs. 10–13(c), 10–13(d), and 10–13(e). Additionally, CNN, Inception-ResNet, and CRF-CNN need to extract training samples from the ground truth and train the convolutional network, while HOTCRF-PCANet just extracts fake samples from the current segmentation to train

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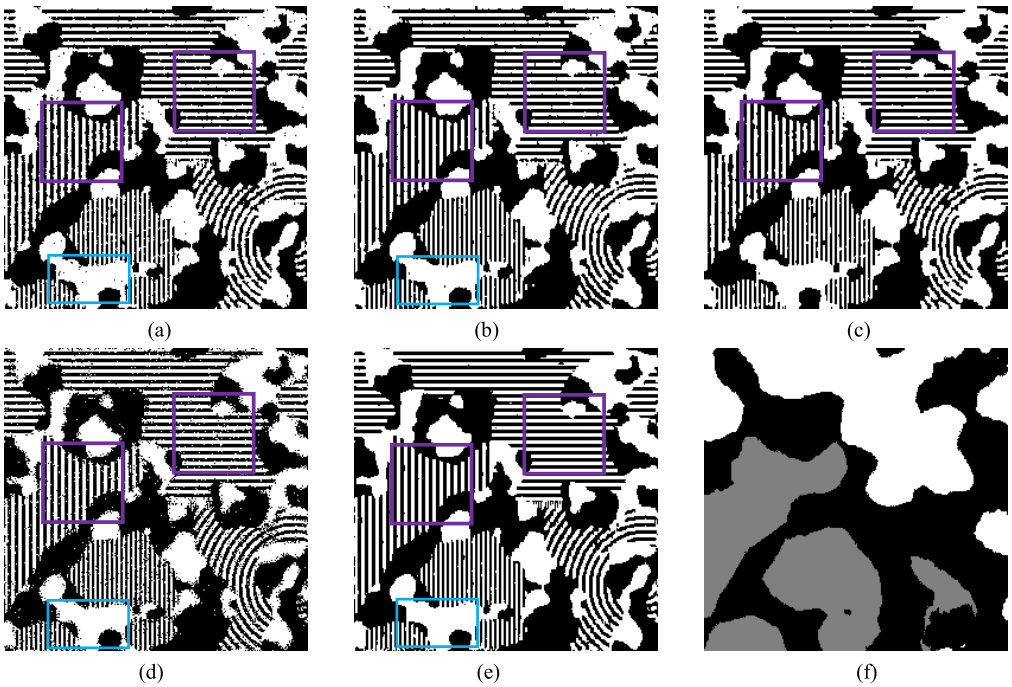

Fig. 11. Segmentation results of simulated image with ENLs being 4-look. (a) CNN. (b) Inception-ResNet. (c) CRF-CNN. (d) HN-TMF. (e) HOTCF-PCANet. (f) Estimated U.

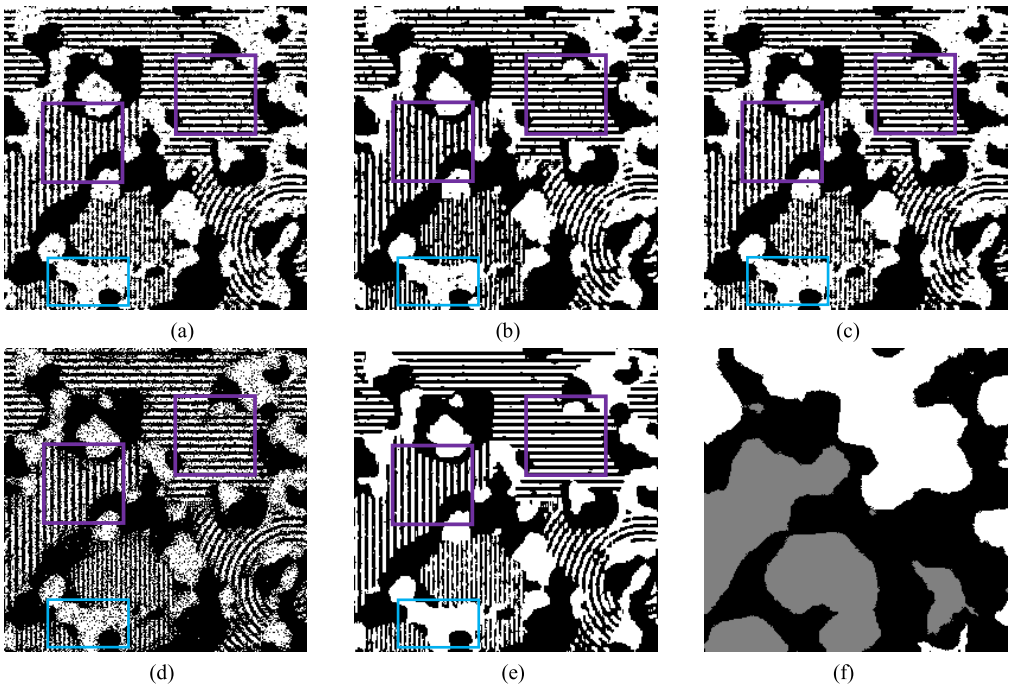

Fig. 12. Segmentation results of simulated image with ENLs being 2-look. (a) CNN. (b) Inception-ResNet. (c) CRF-CNN. (d) HN-TMF. (e) HOTCRF-PCANet. (f) Estimated U.

the UPCANet and construct the UPCANet-based unary potential. Consequently, HOTCRF-PCANet is more suitable for unsupervised segmentation in practice.

F. Experiment and Discussion on SAR Image Segmentation

Real SAR images are shown in Fig. 14 case1-4(a), respectively. The segmentation results by CNN [21], Inception-ResNet [24], CRF-CNN [25], HN-TMF [16] and our proposed HOTCRF-PCANet are shown in Fig. 14 case1-4(b), Fig. 14 case1-4(c), Fig. 14 case1-4(d), Fig. 14 case1-4(e), and Fig. 14 case1-4(f), respectively. For this set of nonstationary SAR images, the auxiliary field is used to regulate two label structure patterns that are weak-structured and intense-structured patterns. The segmentation of the label structure pattern, denoted by the auxiliary field U, is given in Fig. 15 in which black and white denote the weak-structured and intense-structured patterns, respectively.

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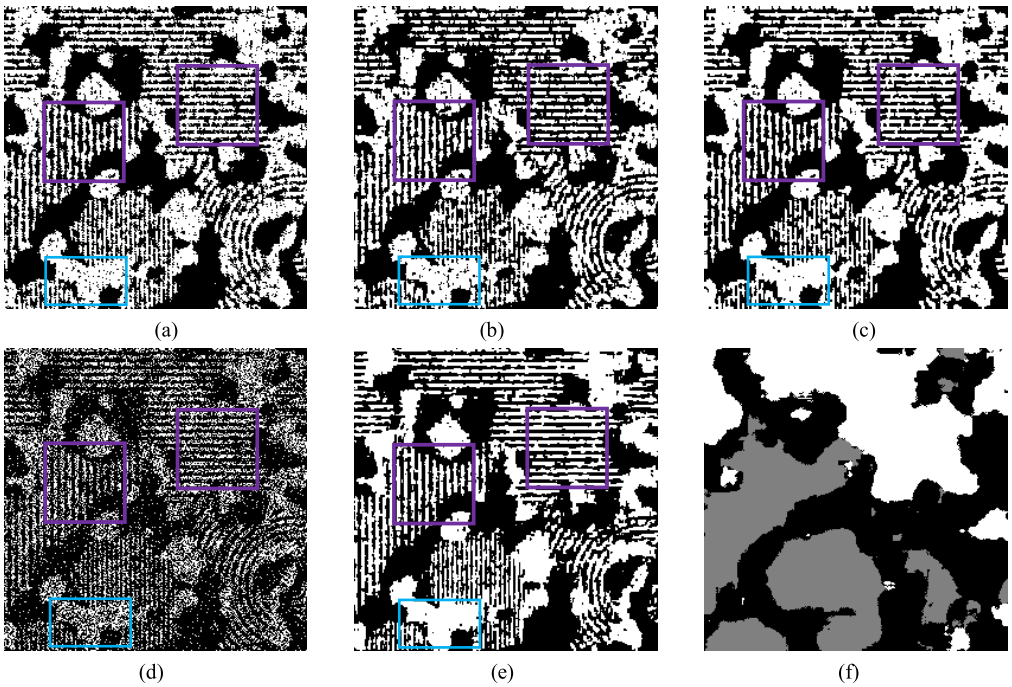

Fig. 13. Segmentation results of simulated image with ENLs being 1-look. (a) CNN. (b) Inception-ResNet. (c) CRF-CNN. (d) HN-TMF. (e) HOTCRF-PCANet. (f) Estimated U.

Under the guidance of this auxiliary field, HOTCRF-PCANet uses the high-order DQNMF potential to model the multidimensional label interactions for pixels. In addition, HOTCRF-PCANet enforces the regions' labeling consistency for pixels within the weak-structured region. According to the segmentation results shown in Fig. 14, the proposed HOTCRF-PCANet can obviously obtain better region homogeneities than these two popular deep learning models, CNN and Inception-ResNet, such as the segmentation of airport shown in the purple panes in Fig. 14 case1-2(b), Fig. 14 case1-2(c), and Fig. 14 case1-2(f), and the segmentation of vegetable zone shown in the purple panes in Fig. 14 case3(b), Fig. 14 case3(c), and Fig. 14 case3(f). Moreover, compared to the results by CNN and Inception-ResNet, more accurate edge locations can be obtained by HOTCRF-PCANet, such as the segmentation of vegetable zone shown in the purple panes in Fig. 14 case3(b), Fig. 14 case3(c), and Fig. 14 case3(f). And HOTCRF-PCANet is more robust against speckle. The reason is that besides taking advantage of the representation-learning ability of UPCANet, HOTCRF-PCANet effectively incorporates the high-order spatial label interaction into the segmentation process. Concretely speaking, the multidimensional label interactions can be captured effectively by the high-order DQNMF potential, and the regions' labeling consistency for pixels within the weak-structured region is enforced by the POE potential of HOTCRF-PCANet.

CRF-CNN, which integrates convolution network with CRF, can take the label interaction into account and obtain better region homogeneity than CNN, as shown in Fig. 14 case1-4(b) and Fig. 14 case1-4(d). But the label interaction is limited within 4-neighborhood system and modeled by the pairwise cliques. In addition, the nonstationarity of the label structure pattern is not considered by CRF-CNN. Consequently, the label structure information cannot be captured effectively by CRF-CNN. Then in the results by CRF-CNN, the image structures are not well preserved and the edge locations of the segmentation are not satisfying. By comparison, HOTCRF-PCANet utilizes an auxiliary field to capture the different kinds of label structures in SAR images, and models the multidimensional label interactions by the proposed high-order DQNMF potential. Then, HOTCRF-PCANet is able to preserve the image structure better than CRF-CNN and thus achieve more meticulous segmentation results. For instance, the segmentation of the building zone by HOTCRF-PCANet is more accurate, as shown in the blue panes in Fig. 14 case2(d) and Fig. 14 case2(f). Moreover, the edge locations of vegetable zone by HOTCRF-PCANet are more accurate, as shown in the purple panes in Fig. 14 case3(d) and Fig. 14 case3(f). And in the weak-structured region, HOTCRF-PCANet focuses on the label consistency and utilizes a POE potential to capture the high-order label dependencies. Consequently, the region homogeneities of the segmentation results by HOTCRF-PCANet are better than those by CRF-CNN, as shown in the purple panes in Fig. 14 case1-4(d) and Fig. 14 case1-4(f). Finally, note that CRF-CNN needs

TABLE VII
COMPARISON OF THE COMPUTATION TIME (SECONDS)

Image	CNN	Inception-ResNet	CRF-CNN	HN-TMF	Proposed
Real SAR1	176	402	308	298	684
Real SAR2	147	367	256	251	566
Real SAR3	131	400	224	207	610
Real SAR4	219	492	374	332	987

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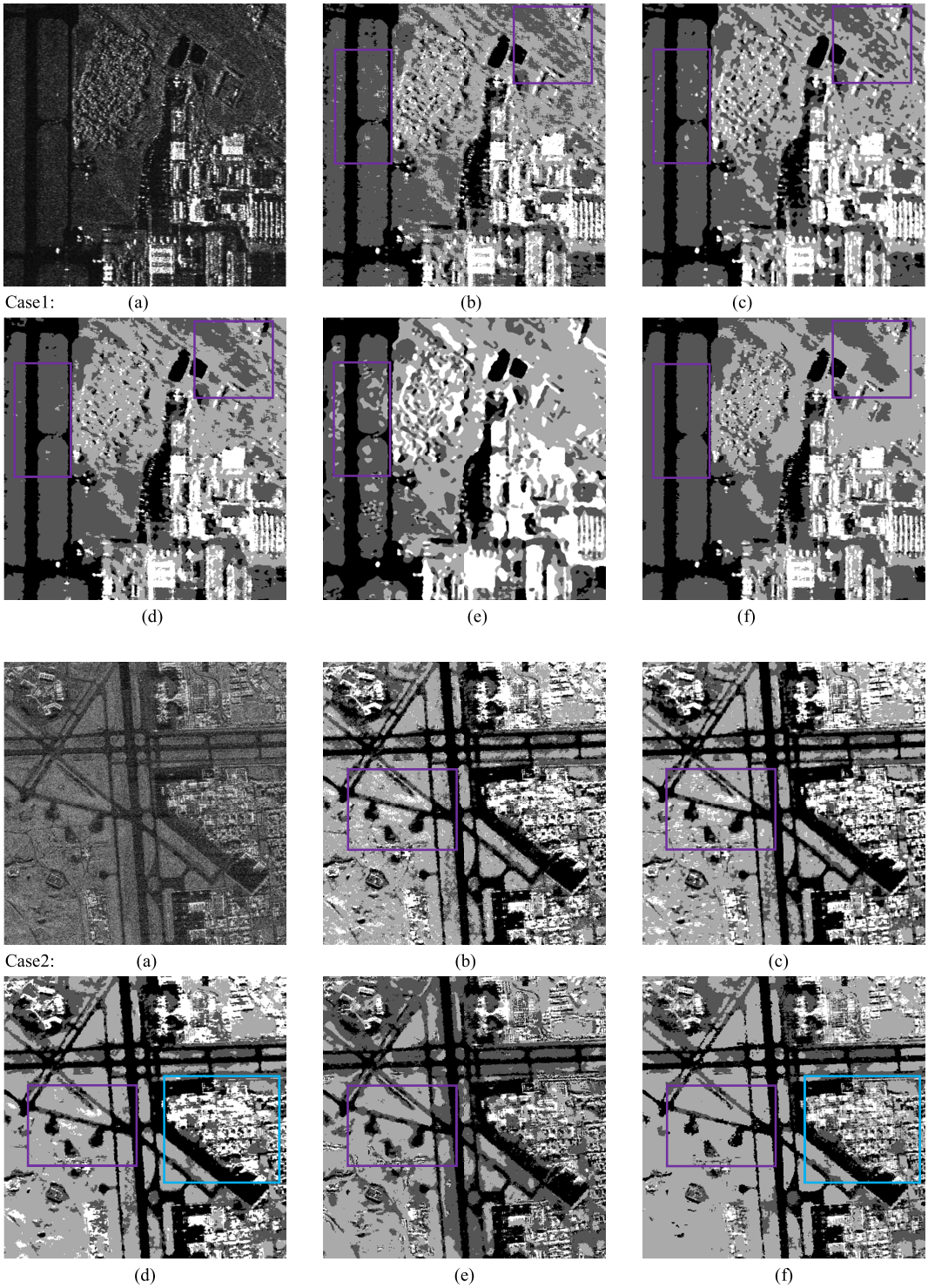

Fig. 14. Segmentation results of real SAR images. Case1-4(a) SAR images. Case1-4(b) Segmentation based on CNN. Case1-4(c) Segmentation based on Inception-ResNet. Case1-4(d) Segmentation based on CRF-CNN. Case1-4(e) Segmentation based on HN-TMF. Case1-4(f) Segmentation based on HOTCRF-PCANet.

200 samples for each class manually according to the prior information, while 100 fake samples for each class are needed by the proposed HOTCRF-PCANet to train the UPCANet in the segmentation. Thus, the proposed HOTCRF-PCANet is more suitable for unsupervised segmentation.

The experimental results also demonstrate the superiority of HOTCRF-PCANet over HN-TMF model for SAR image segmentation. Because HN-TMF model uses statistics rather than deep features, and does not model the multidimensional label structure explicitly, there are some mis-segmentations in the results by HN-TMF, such as the segmentation of airport runway in the purple pane in Fig. 14 case1(e) and the segmentation of vegetable shown in the purple panes in Fig. 14 case3-4(e). In addition, to ensure the

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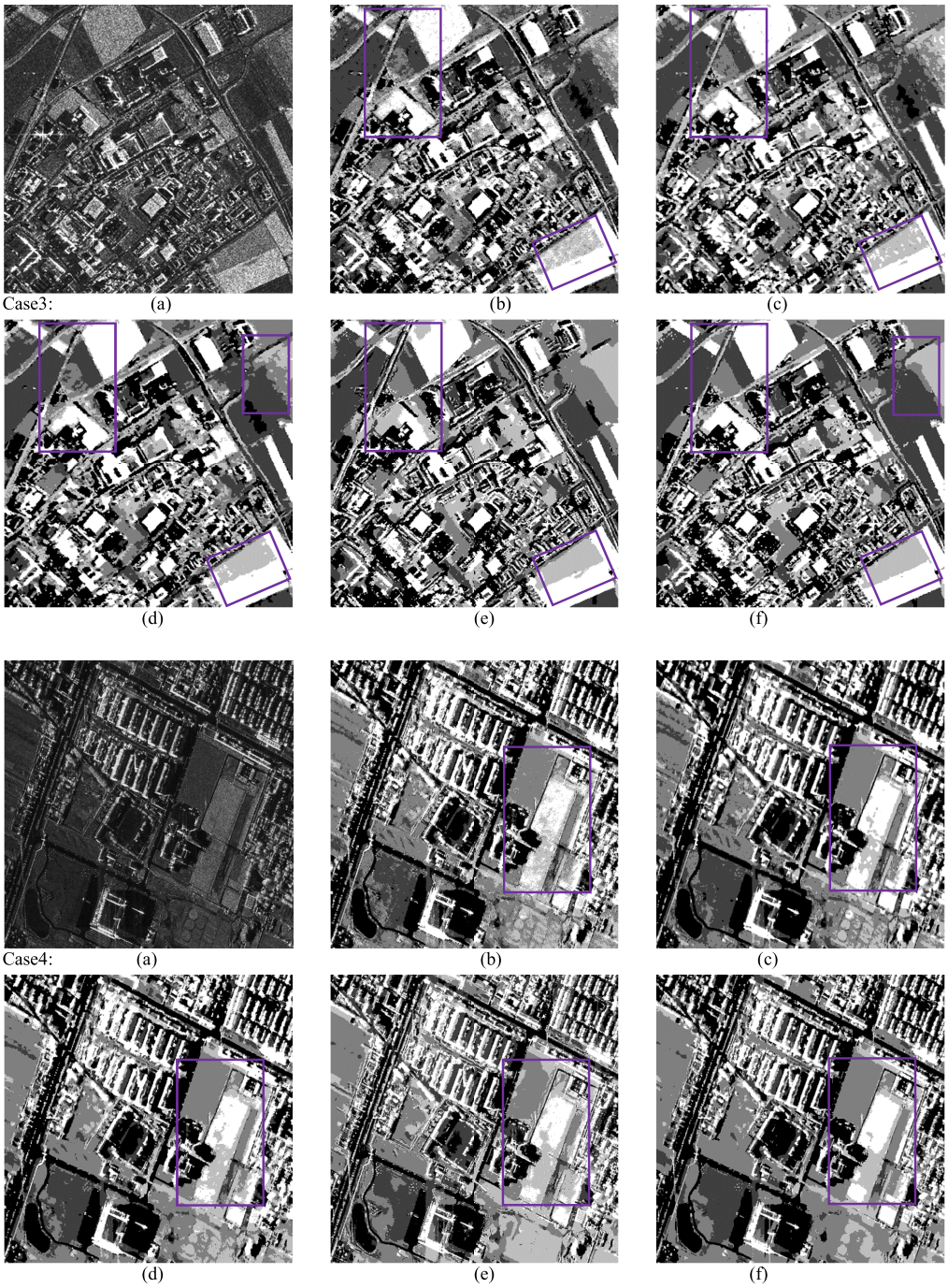

Fig. 14. *(Continued)* Segmentation results of real SAR images. Case1-4(a) SAR images. Case1-4(b) Segmentation based on CNN. Case1-4(c) Segmentation based on Inception-ResNet. Case1-4(d) Segmentation based on CRF-CNN. Case1-4(e) Segmentation based on HN-TMF. Case1-4(f) Segmentation based on HOTCRF-PCANet.

region homogeneity, HN-TMF sometimes obtains over-segmentations, and thus the image structures are not preserved well, such as the segmentation of the buildings shown in Fig. 14 case1(e). By comparison, HOTCRF-PCANet can guarantee the region homogeneity of segmentation and preserve the image structure better, such as the segmentation of the buildings shown in Fig. 14 case1(f). Moreover, some mis-segmentations of HN-TMF model can be obviously corrected by HOTCRF-PCANet, as shown in the purple panes in Fig. 14 case1(f) and Fig. 14 case3-4(f). The reason is as follows: 1) By the integration of deep learning network into the proposed HOTCRF framework, HOTCRF-PCANet can achieve more accurate segmentation using the deep features; 2) The proposed high-order DQNMF potential explicitly models the high-order label interactions through a zigzag scan and thus can effectively capture the label structure information in the segmentation.

The estimated auxiliary field U of each SAR image is shown in Fig. 15. As shown in Fig. 15, the obtained auxiliary field effectively segments the label structure pattern of each image,

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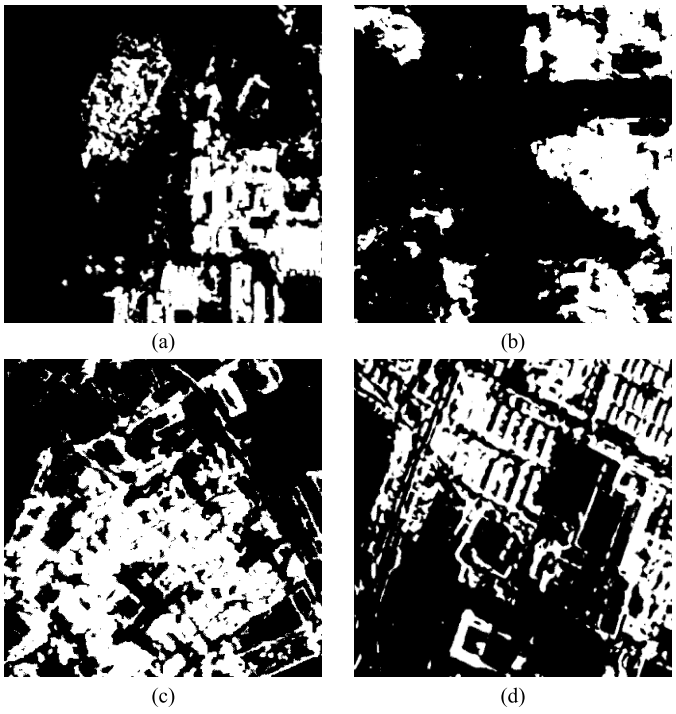

Fig. 15. Estimated U of real SAR images (black and white regions denote the weak-structured and intense-structured patterns, respectively). (a) Case1. (b) Case2. c) Case3. (d) Case4.

in which black and white regions denote the weak-structured and intense-structured patterns, respectively. Actually, under the guidance of the auxiliary field, HOTCRF-PCANet takes the different label structure patterns of nonstationary image into account, and thus models the label interactions by (10)–(12). In addition, it can effectively enforce the regions' labeling consistency for pixels within the weak-structured region by using the POE potential. Thus, HOTCRF-PCANet can capture image structure more accurately in the segmentation.

The computation time of HOTCRF-PCANet and the reference models for real SAR images are listed in Table VII. The segmentation is executed on Intel Core i7, 4 GHz. In practice, the computation burden of HOTCRF-PCANet is acceptable compared to other models, with the segmentation performance of HOTCRF-PCANet being the best generally.

VII. Conclusion

In this article, we propose a HOTCRF-PCANet for the unsupervised segmentation of nonstationary SAR images. The proposed HOTCRF-PCANet introduces an auxiliary field to segment the nonstationary image into weak-structured and intense-structured regions and thus explicitly regulate nonstationary label structure patterns. Under the guidance of this auxiliary field, HOTCRF-PCANet defines a high-order DQNMF potential and thus effectively models multidimensional label interactions. In addition, for the pixels within the weak-structured region, HOTCRF-PCANet enforces the regions' labeling consistency in the segmentation by utilizing an auxiliary-field-guided POE potential. Moreover, we modify PCANet into UPCANet for unsupervised segmentation, and integrate this deep learning network into the CRF framework, thus effectively enhancing the representation-learning ability of HOTCRF-PCANet. In this way, HOTCRF-PCANet successfully combines the high-order label-interaction modeling power of random fields model with the representation-learning ability of deep learning networks. Consequently, HOTCRF-PCANet achieves better structure preservation, edge location, and region homogeneity than the reference model proposed recently in the segmentation of simulated images and real SAR images. As future work, we plan to study the introduction of generalized principal component analysis in the deep learning network which can deal with SAR data with non-Gaussian statistics, and thus enhance the robustness against speckle.

Appendix

Here, we present the derivation of the multidimensional probability based on DQNMF in detail. First, for simplicity, we have the following notation. $\mathbf{V}_{i',j'} = (\mathbf{X}_{i',j'}, \mathbf{U}_{i',j'})$ denote the pairwise hidden random variables, and $v_{i',j'} = (x_{i',j'}, u_{i',j'})$ denote their realizations. Then, Theorem 1 is given as follows. For all i, j with $1 \le i \le n, 1 \le j \le n$, if

$$\begin{aligned} p\left(v_{i,j}/Z_{i,j}^{ul}\right) &= p(v_{i,j}/v_{i-1,j}, v_{i,j-1}) \\ p\left(v_{i,j}/Z_{i,j}^{dr}\right) &= p(v_{i,j}/v_{i+1,j}, v_{i,j+1}) \\ p\left(v_{i,j}/Z_{i,j}^{ur}\right) &= p(v_{i,j}/v_{i-1,j}, v_{i,j+1}) \\ p\left(v_{i,j}/Z_{i,j}^{dl}\right) &= p(v_{i,j}/v_{i+1,j}, v_{i,j-1}) \end{aligned} \tag{28}$$

where $Z_{i,j}^{ul} = \{(x_{i'',j''}, u_{i'',j''}) | i'' < i \text{ or } j'' < j\}$, $Z_{i,j}^{dr} = \{(x_{i'',j''}, u_{i'',j''}) | i'' > i \text{ or } j'' > j\}$, $Z_{i,j}^{ur} = \{(x_{i'',j''}, u_{i'',j''}) | i'' < i \text{ or } j'' > j\}$, $Z_{i,j}^{dl} = \{(x_{i'',j''}, u_{i'',j''}) | i'' > i \text{ or } j'' < j\}$ are the predecessors of $(x_{i,j}, u_{i,j})$, the multidimensional probability of the multidimensional hidden variables in the high-order neighborhood can be expressed as

$$\begin{aligned} p(\boldsymbol{v}_{m',n}) = \prod_{i=1}^{m'} \prod_{j=1}^{n} & p(v_{2i-1,j} | v_{2i-2,j}, v_{2i-1,j-1}) \\ & \cdot p(v_{2i,n-j+1} | v_{2i-1,n-j+1}, v_{2i,n-j+2}) \end{aligned} \tag{29}$$

where $\boldsymbol{v}_{m',n} = \{v_{2i-1,1}, \ldots, v_{2i-1,n}, v_{2i,1}, \ldots, v_{2i,n}\}_{1 \le i \le m'}$ and $m' = (n-1)/2 + 1$.

Motivated by the derivation in [49], we present the proof of Theorem 1 as follows.

Proof: Assuming Theorem 1 holds for $\boldsymbol{v}_{m',n}$, we show that it holds for $\boldsymbol{v}_{m'+1,n}$, and induction on n follows by symmetry of the expressions. Now, we have

$$\begin{aligned} p(\boldsymbol{v}_{m'+1,n}) &= p(\boldsymbol{v}_{m',n}, v_{2(m'+1)-1,n}, \ldots, v_{2(m'+1)-1,1}, \\ &\qquad v_{2(m'+1),1}, \ldots, v_{2(m'+1),n}) \\ &= p(v_{2(m'+1)-1,n}, \ldots, v_{2(m'+1)-1,1}, \\ &\qquad v_{2(m'+1),1}, \ldots, v_{2(m'+1),n} | \boldsymbol{v}_{m',n}) p(\boldsymbol{v}_{m',n}) \\ &= p(v_{2(m'+1)-1,n}, \ldots, v_{2(m'+1)-1,1} | \boldsymbol{v}_{m',n}) \\ &\quad \cdot p(v_{2(m'+1),1}, \ldots, v_{2(m'+1),n} | v_{2(m'+1)-1,n}, \ldots, \\ &\qquad v_{2(m'+1)-1,1}, \boldsymbol{v}_{m',n}) p(\boldsymbol{v}_{m',n}) \end{aligned} \tag{30}$$

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and then $p(v_{2(m'+1),1},\ldots,v_{2(m'+1),n}|v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})$ can be expressed as

$$\begin{aligned}&p(v_{2(m'+1),1},\ldots,v_{2(m'+1),n}|v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\\&=p(v_{2(m'+1),1}|v_{2(m'+1),2},\ldots,v_{2(m'+1),n},v_{2(m'+1)-1,n},\\&\quad v_{2(m'+1)-1,n-1},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\\&\quad\cdot p(v_{2(m'+1),2}|v_{2(m'+1),3},\ldots,v_{2(m'+1),n},v_{2(m'+1)-1,n},\\&\quad v_{2(m'+1)-1,n-1},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\cdots\\&\quad\cdot p(v_{2(m'+1),n}|v_{2(m'+1)-1,n},v_{2(m'+1)-1,n-1},\ldots,\\&\quad v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n}).\end{aligned}\tag{31}$$

By using the condition $p(v_{i,j}/Z^{ur}_{i,j})=p(v_{i,j}/v_{i-1,j},v_{i,j+1})$ given by (28), we have

$$\begin{aligned}&p(v_{2(m'+1),1},\ldots,v_{2(m'+1),n}|v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\\&=p(v_{2(m'+1),n-n+1}|v_{2(m'+1)-1,n-n+1},v_{2(m'+1),n-n+2})\\&\quad\cdot p(v_{2(m'+1),n-(n-1)+1}|v_{2(m'+1)-1,n-(n-1)+1},\\&\quad v_{2(m'+1),n-(n-1)+2})\cdots\\&\quad\cdot p(v_{2(m'+1),n-1+1}|v_{2(m'+1)-1,n-1+1},v_{2(m'+1),n-1+2}).\end{aligned}\tag{32}$$

And $p(v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1}|\boldsymbol{v}_{m',n})$ can be expressed as

$$\begin{aligned}&p(v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1}|\boldsymbol{v}_{m',n})\\&\quad=p(v_{2(m'+1)-1,n}|v_{2(m'+1)-1,n-1},\ldots,v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\\&\quad\cdot p(v_{2(m'+1)-1,n-1}|v_{2(m'+1)-1,n-2},\ldots,\\&\quad v_{2(m'+1)-1,1},\boldsymbol{v}_{m',n})\cdots\\&\quad\cdot p(v_{2(m'+1)-1,1}|\boldsymbol{v}_{m',n}).\end{aligned}\tag{33}$$

By using the condition $p(v_{i,j}/Z^{ul}_{i,j})=p(v_{i,j}/v_{i-1,j},v_{i,j-1})$ given by (28), we have

$$\begin{aligned}&p(v_{2(m'+1)-1,n},\ldots,v_{2(m'+1)-1,1}|\boldsymbol{v}_{m',n})\\&\quad=p(v_{2(m'+1)-1,n}|v_{2(m'+1)-1,n-1},v_{2(m'+1)-2,n})\\&\quad\cdot p(v_{2(m'+1)-1,n-1}|v_{2(m'+1)-1,n-2},v_{2(m'+1)-2,n-1})\cdots\\&\quad\cdot p(v_{2(m'+1)-1,1}|v_{2(m'+1)-1,0},v_{2(m'+1)-2,1}).\end{aligned}\tag{34}$$

Then, by using (30), (32), and (34), $p(\boldsymbol{v}_{m'+1,n})$ can be expressed as

$$\begin{aligned}&p(\boldsymbol{v}_{m'+1,n})\\&\quad=p(\boldsymbol{v}_{m',n})\\&\quad\cdot[p(v_{2(m'+1)-1,n}|v_{2(m'+1)-1,n-1},v_{2(m'+1)-2,n})\\&\quad\cdot p(v_{2(m'+1)-1,n-1}|v_{2(m'+1)-1,n-2},v_{2(m'+1)-2,n-1})\cdots\\&\quad p(v_{2(m'+1)-1,1}|v_{2(m'+1)-1,0},v_{2(m'+1)-2,1})]\\&\quad\cdot[p(v_{2(m'+1),n-n+1}|v_{2(m'+1)-1,n-n+1},v_{2(m'+1),n-n+2})\\&\quad\cdot p(v_{2(m'+1),n-(n-1)+1}|v_{2(m'+1)-1,n-(n-1)+1},\\&\quad v_{2(m'+1),n-(n-1)+2})\cdots\\&\quad p(v_{2(m'+1),n-1+1}|v_{2(m'+1)-1,n-1+1},v_{2(m'+1),n-1+2})]\\&\quad=p(\boldsymbol{v}_{m',n})\cdot\prod_{j=1}^{n}p(v_{2(m'+1)-1,j}|v_{2(m'+1)-1,j-1},v_{2(m'+1)-2,j})\\&\quad\cdot p(v_{2(m'+1),n-j+1}|v_{2(m'+1)-1,n-j+1},v_{2(m'+1),n-j+2}).\end{aligned}\tag{35}$$

Then, by using (35), $p(\boldsymbol{v}_{m'+1,n})$ can be rewritten by

$$\begin{aligned}p(\boldsymbol{v}_{m'+1,n})=&\prod_{i=1}^{m'+1}\prod_{j=1}^{n}p(x_{2i-1,j},u_{2i-1,j}|x_{2i-2,j},u_{2i-2,j},\\&x_{2i-1,j-1},u_{2i-1,j-1})\\&\cdot p(x_{2i,n-j+1},u_{2i,n-j+1}|x_{2i-1,n-j+1},u_{2i-1,n-j+1},\\&x_{2i,n-j+2},u_{2i,n-j+2})\end{aligned}\tag{36}$$

Therefore, Theorem 1 is true and we can derive the multidimensional probability of the multidimensional hidden variables. Note that given the free boundary assumption, the sites on the boundary only interact with neighbors located inside the lattice and are independent of the ones that are outside the lattice.

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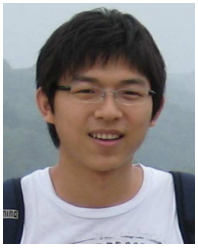
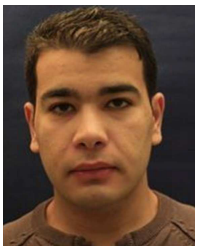


Peng Zhang (Member, IEEE) received the B.S. degree in electronic and information engineering and the M.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 2006, 2009, and 2012, respectively.

He is an Associate Professor with the National Key Laboratory of Radar Signal Processing, Xidian University. His main research interests are SAR image interpretation and statistical learning theory.



Mohamed El Yazid Boudaren received the Ph.D. degree in computer science from the Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algiers, Algeria, and Université Pierre et Marie Curie, Paris, France, in 2014, and the Habilitation degree in computer science from USTHB in 2015.

He is an Associate Professor with the Ecole Militaire Polytechnique, Algiers. His research interests include stochastic processes and artificial intelligence.

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Yinyin Jiang received the B.S. degree in electrical science and technology from Anhui Normal University, Wuhu, China, in 2017. She is pursuing the Ph.D. degree with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

Her primary research interest is polarimetric synthetic aperture radar image classification.

Wanying Song (Member, IEEE) was born in 1988. She received the B.S. degree in electrical science and technology from the Shandong University of Science and Technology, Qingdao, China, in 2012, and the Ph.D. degree in signal and information processing from the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China, in 2018.

Her main research interest is polarimetric synthetic aperture radar image classification.

Beibei Li received the B.S. degree in electrical science and technology from the Shandong University of Science and Technology, Qingdao, China, in 2018. She is pursuing the master's degree with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

Her primary research interests are SAR image analysis and interpretation.

Ming Li was born in 1965. He received the B.S. degree in electrical engineering and the M.S. and Ph.D. degrees in signal processing from Xidian University, Xi'an, China, in 1987, 1990, and 2007, respectively.

In 1987, he joined the Department of Electronic Engineering, Xidian University, where he is a Professor with the National Key Laboratory of Radar Signal Processing. His research interests include adaptive signal processing, detection theory, ultra-wideband, and SAR image processing.

Yan Wu (Member, IEEE) received the B.S. degree in information processing and the M.A. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, China, in 1987, 1998, and 2003, respectively.

From 2003 to 2005, she was a Postdoctoral Fellow with the National Key Laboratory of Radar Signal Processing. Since 2005, she has been a Professor with the Department of Electronic Engineering, Xidian University. She has published more than 60 technical articles. Her broad research interests are remote sensing image analysis and interpretation, data fusion of multi-sensor images, SAR auto target recognition, and statistical learning theory and application.

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